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Acts of Survival

Exploring trust through a multiplayer analog game

Department of Game Design

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Abstract

Trust has been explored as an important factor in game research, expressed across various multiplayer games through game mechanics. However, less attention was paid to how these mechanics specifically affected it. This study investigates how trust can be modelled and explored in a multiplayer analog game through its mechanics with a focus on reflective game design.

Using an iterative research through design approach, this study developed and evaluated a two-player analog game, *"Act of Survival"*, across three playtests and design iterations. The data were gathered through observations and semi-structured interviews and analysed using thematic analysis to identify patterns in players' trust-related behaviour.

The findings indicate that trust is a process shaped by interactions among players in situations where dependency, uncertainty, vulnerability, risk, and the possibility of betrayal exist. The findings highlight the importance of mechanics that create dependency between players, incomplete information, the possibility of deception, and player agency to make trust-based decisions. Together, these factors allowed players to actively negotiate on how much trust they gave the other players and made it possible to reflect on trust as part of gameplay.

This study contributes to the understanding of how trust can be modelled and explored in a game system through its mechanics. By showing how specific mechanics can model different dimensions of trust, the findings provide insights for designers seeking to create trust-based experiences.

Abstract **1**
1. Introduction **4**
2. Background **5**
2.1. Literature Review 5
2.2. Trust in Games 7
2.3. Theoretical Framework 8
3. Game overview **10**
4. Methodology **15**
4.1. Research Through Design 15
4.2. Data Collection and Analysis 16
4.2.1. Playtesting and Data Collection 16
4.2.2. Data Analysis 17
4.3. Positionality 17
4.4. Ethical Considerations 18
4.5. Limitations 18
5. Results and Analysis **18**
5.1. Iteration 1 of Act of Survival: First Prototype and Initial Playtest Session 19
5.1.1. Prototype 1 19
5.1.2. Design choices prototype 1 22
5.1.3. Playtest Session 1 23
5.1.4. Observations 23
5.1.5. Interview Results 26
5.1.6. Findings of Prototype 1 28
5.2. Iteration 2 of Act of Survival: Second Prototype and Second Playtest Session 30
5.2.1. Prototype 2 30
5.2.2. Design choices prototype 2 30
5.2.3. Playtest Session 2 33
5.2.4. Observations 33
5.2.5. Interview Results 36
5.2.6. Findings of Prototype 2 38
5.3. Iteration 3 of Act of Survival: Third Prototype and Final Playtest Session 39
5.3.1. Prototype 3 39
5.3.2. Design choices prototype 3 41
5.3.3. Playtest Session 3 42
5.3.4. Observations 42
5.3.5. Interview Results 44
5.3.6. Findings of Prototype 3 48
6. Discussion **49**
6.1. Themes 49

6.2. Mechanics and Design Implications..... 54
6.3. Answering the Research Question..... 56
7. Conclusion.....57
8. References..... 59
9. Ludography.....60
10. Appendices..... 61
10.1. Consent form..... 61
10.2. Observations.....64
10.3. Semi structured interviews..... 65
10.4. Rules and Narrative introduction..... 65
10.5. Observation and Interview Data.....67
10.6. Gameplay Video.....67

Keywords: Trust, reflection, game mechanics, social exchange

1. Introduction

Games can be more than just systems that were created solely for entertainment; they can also be about something deeper. According to Rusch and Weise (2008), their previous research on deep and meaningful games shows that games can tackle complex concepts and human experiences from a medium-specific perspective, allowing players to engage with thought-provoking and insightful experiences through interaction and play. Through mechanics, systems, and player decision-making, games can function as experiential metaphors that encourage reflection rather than simply conveying meaning through narrative alone.

This thesis originated from our previous course, Deep Games, where the project started as an exploration of how complex social experiences, such as trust, could be metaphorically expressed and designed through game mechanics. Inspired by ideas of deep games and the work of Rusch and Weise (2008), we set out to design a game that allows the players to explore their relationship to trust and reflect on those experiences.

Rather than designing trust as a predefined system or fixed condition, the project focuses on how trust can emerge through player interaction, negotiation, and decision-making. To research this, an iterative research through design process was implemented in which a two-player game prototype was developed and evaluated across three playtesting sessions.

Based on this design process and our aim, to guide our project, we formulated this research question:

> *How can we create a game system that creates a space for players to explore and reflect on their relationship with trust through the interactions with another player?*

Our research and insights from the design process could be useful for game designers or researchers interested in analog games, helping them create game systems that could build the dynamics of complex concepts such as trust. We also believe that game researchers interested in how games may affect relationship dynamics can take inspiration from our research and build on it.

2. Background

According to Rotter (1980, para. 2), interpersonal trust affects all kinds of relationships, including relationships between governments, buyers, sellers, parents and children. She also argues that unwarranted distrust and gullibility can lead to serious negative consequences, such as deteriorated relationships or missed opportunities. We want to dive deeper into how game mechanics can affect the relationship between players and create space for reflection on trust through interactions with another player.

According to Koster (2018), the trust spectrum is a design-focused framework that examines how game design elements influence trust between players. The author describes trust as a spectrum, in which different games require varying levels of dependence between players. Koster discusses how trust in games is shaped by mechanics such as player interdependence, collaboration, shared versus individual goals, and the distribution of information among players. According to Koster, there are four ranges of trust in the trust spectrum: no trust, low trust, medium trust and high trust. Games that require stronger cooperation are placed higher on the trust spectrum. Koster's (2018) work provided us with insight into how design elements influence trust, which is why we focused on implementing dependency, collaboration, limited information, shared and individual goals, and unique resources to land in the high-trust range of the spectrum.

According to Gottman (2011), trust is built in small moments, which he calls "sliding door moments". In any interaction, we may connect with our partner or turn away from them. You build trust when you help another rather than focusing on yourself; it could be as simple as listening to your partner after having a bad day instead of doing what you originally planned. We drew inspiration from Gottman (2011), in which we tried to implement smaller moments early in the game when the mechanics created opportunities for players to help each other.

2.1. Literature Review

In this section, we provide definitions of trust from different fields, such as psychology, sociology, and organisational psychology, and present prior research in game studies on trust.

The concept of trust has received significant recognition across various domains and communities, and it is considered a major factor in decision-making that can further build relationships (Cho et al., 2015, para. 1). The author provides a definition of trust, which is;

"Trust is the willingness of the trustor (evaluator) to take risk based on a subjective belief that a trustee (evaluatee) will exhibit reliable behaviour to maximise the trustor's interest under uncertainty (for example, ambiguity due to conflicting evidence and/or ignorance caused by complete lack of evidence) of a given situation" (Cho et al., 2015, p. 5). Based on this definition, our goal was to create scenarios in which players had to take risks under uncertainty.

Rotter (1980, para.1) examines both the positive and negative consequences of being a high or low-trust individual in interpersonal trust in social situations, specifically when interacting with ordinary people. In the context of social learning theory, the author defines interpersonal trust as a general expectancy that the words, promises, or statements of other individuals can be relied upon. Expectancy in different situations is determined not only by specific experiences in those situations but also by other situations the individual perceives as similar to previous ones (Rotter, 1980, pp. 1-2). Inspired by Rotter's (1980) definition of interpersonal trust, we implemented a mechanic in which players cannot show their abilities, but can talk freely about them. As a result, the players had to rely on the other player's word.

Mayer (1995, p. 712) defines trust as a party's willingness to expose themselves to another party's actions, with the expectation that the other will perform a particular action that could benefit the trustor, regardless of the trustor's ability to monitor that other party. Making oneself vulnerable is about taking risks. Trust is not simply taking risks, but the willingness to expose oneself to risk. However, that willingness to take risks depends on three perceived qualities of the other party: ability, benevolence and integrity. From Mayer's (1995) study on trust, we take away that trust depends on risk and vulnerability, so we considered these factors as they emerged while building the mechanics.

Trust is a particular level of subjective probability that a trustor assigns to the trustee's performance of an action that could be beneficial or at least not harmful, before the trustor can monitor that action. For trust to exist, there must be the possibility of exit, betrayal, defection. If the trustee's actions are heavily constrained, that could limit the trustor's freedom, restricting the field of actions that must be guessed and performed, in turn limiting or removing uncertainty. (Gambetta, 1988, para. 12-15). We aimed to use Gambetta's (1988) work on trust by incorporating possibilities of betrayal without restricting the player's actions, thereby removing uncertainty.

Inspired by the previous literature, we believe that trust is an emotional decision to rely on another individual in vulnerable and challenging situations, in which one willingly exposes oneself to risk, with the understanding that outcomes are uncertain and that betrayal is possible. In an interaction between two individuals, the expectation that you will fulfil your statements or promises is also a crucial aspect of trust. According to how we define trust, it is constituted by risk, vulnerability, dependency, uncertainty, and possible betrayal. We base our definition of trust on the work of Gambetta (1988), Mayer (1995), Rotter (1980), and Cho et al. (2015), who examine what trust is and what constitutes it.

In their study, Fehlhaber and El-Awad (2024, pp. 1-4) explored how trust can be built between strangers in a hostile, competitive online gaming environment. In total, they ran nine games involving 101 participants aged 14-59, with 4365.68 hours of playtime. The game they used to conduct their research is an adaptation of the social deduction game Tank Turn Tactics. In the game, trust and mistrust can emerge as players perform various competitive, cooperative, or self-referential tasks in an effort to be the last survivor.

During the study, there was no Nash equilibrium (no player can improve their payoff by changing their strategy). Instead, the players were given the freedom to choose between trust and defection throughout the game. The participants could communicate freely without restrictions or having to disclose their anonymity. (Fehlhaber & El-Awad, 2024, p. 3)

Even though conditions were put in place for distrust to happen, trust between the participants emerged. The acts of trust occurred most often when participants were in pairs or trios. The results showed that trust is less dependent on the platform than on social interactions, such as Mutual trust and alliance formation, betrayal and deception, resource distribution and cooperation, Power Dynamics between players and the impact of time constraints. (Fehlhaber & El-Awad, 2024, pp. 29-34). Similar to Fehlhaber and El-Awad (2024), we focused on social interactions, mutual trust, betrayal/deception and resource distribution. However, our research focuses on an analog environment that promotes exploration and reflections on trust rather than building it.

Depping and Mandryk (2017, para. 1) examine two collaborative mechanisms, cooperation and interdependence, and test how they can foster social closeness between two paired strangers in online games. Their findings indicate that mechanics requiring players to rely on one another and coordinate their actions can significantly increase feelings of connection and positive social interaction, even between strangers. This study is relevant to our work as it highlights how interdependent mechanics can shape player relationships by structuring reliance and interaction. In the context of this thesis, such mechanics are important for creating conditions in which trust can emerge, as dependency and cooperation form the foundation for trust-based decision making (Depping & Mandryk, 2017, pp. 451-452.).

2.2. Trust in Games

Multiple games have mechanics related to trust, cooperation and player interactions. These games demonstrate how mechanics such as dependency, hidden information and individual goals can create tension and conflict between players during gameplay.

The game *Among Us* (2018) revolves around hidden roles and deception as central mechanics, requiring players to constantly evaluate others' intentions and trustworthiness under conditions of uncertainty. As players must interpret social behaviour and decide whom to believe while still managing the risk of being deceived or betrayed, trust becomes fragile and negotiable. Through its system, the game displays how uncertainty and hidden information can create social tension and conflict between players.

In the game *Keep Talking and Nobody Explodes (2015),* players must rely on communication and asymmetric information, such as symbols or codes*,* to succeed. Player one has access to the bomb, while player two has instructions on how to defuse it, which forces the players to rely on each other through information sharing. This game demonstrates how dependency and communication can promote trust.

Dead of Winter (2014) combines cooperation, hidden information, and individual objectives, creating tensions among players as they must balance collective survival with personal self-interest. Players need to share resources to survive, yet they do not know each other's individual goals, whether someone is committed to the group, or who is going to betray them. The game sets up situations in which trust, deception, suspicion, and negotiation emerge through gameplay.

A Way Out (2018) is an action-adventure game that focuses on enforced co-operation between two players through shared goals and coordinated action rather than unique resources. The gameplay is asymmetric, with distinct mechanics and viewpoints for players. Hence, the progression of the game depends on both players relying on each other through collaborative work, synchronised decisions, and actions. The gameplay demonstrates how co-operation and interdependence can foster player relationships and encourage communication among players.

While these games take different approaches to conflict and player interactions, there are few opportunities to explore trust, as the systems force players to play in a certain way to win. This provides us with an opportunity to research how game systems can be designed to allow players to actively explore all aspects of trust and reflect on trust through gameplay.

2.3. Theoretical Framework

With this theoretical framework, we aim to provide a foundation for designing and analysing the ways in which players can explore trust in our analog game. To do this, we utilise Reflective game design (Khaled, 2018), Social exchange theory from Homans (1958) and Molm (2010), How trust can be modelled (Cho et al., 2015) and the MDA (Mechanics, Dynamics and Aesthetics) framework (Hunicke et al., 2004).

Reflective game design is used to understand how it can encourage players to reflect on their experiences. Social exchange theory contributes to knowledge of social behaviour between individuals, in which the exchange of material and symbolic goods takes place, and how the reciprocity of these exchanges fosters bonds between individuals. Trust is not universal but varies across domains and can be modelled and assessed by understanding its attributes. The MDA framework is used to analyse how mechanics, dynamics and aesthetics are connected and mutually built on one another.

> Reflection is a mental process that occurs when individuals encounter situations they cannot address with their previous experiences or solutions. (Khaled, 2018, para. 1).

In our opinion, there is great value in designing a game that allows players to reflect on complex social concepts. Trust is often something people experience every day with other people, everything from trusting a co-worker to do their job, having someone trust you to follow through on your promises, and generally determining if people are genuine or driven by self-interest. Situations like these can create good relationships but also conflict between people, especially when trust can be broken. By having players reflect on their experiences in

regards to trust with another person, we believe people can learn more about themselves, others, and what trust truly means.

One of our primary goals is for players to reflect on their experience and the decisions they made while playing our game regarding trust. By utilising reflective game design, we hope to achieve that. According to Khaled (2018, pp. 1-2), games are a valuable tool for triggering and supporting reflection. However, she argues that conventional game design goes against reflection when it prioritises design for every player and immersion. Instead, Khaled (2018, pp. 20-21) proposes Reflective game design, which promotes questions over answers, meaning it is more about creating spaces and opportunities for players to explore different possibilities rather than providing them with the correct solutions. She argues that meaningful questions are more important than answers.

We aim for our game to run on a core design that relies on the exchange of resources and decisions between players, using these mechanics to enhance the probability of success for both players. This exchange of resources connects closely with social exchange theory.

Social exchange theory originates from Homans (1958, p. 606) proposal of social behaviour as a system of exchange, in which individuals trade material and non-material goods, such as symbols of approval or status, seeking rewards that outweigh possible costs. The author Homans (1958, para. 1) explicitly positions exchange theory as interdisciplinary, aiming to bridge behavioural psychology, economics, influence dynamics, and small-group structure within a unified exchange framework. Drawing on behavioural psychology, the author Homans (1958, pp. 598-603) treats interaction as governed by behaviours that are reinforced and those that are not. From this perspective, social interaction resembles an economy of actions in which individuals engage in rewarding activities and decline those with recurring costs, seeking to maximise their profit. Homans (1958, p. 599) says that the value of the activity influences the frequency of exchanges, so if a person finds the activity rewarding, they must adjust their behaviour to produce valuable trade with the other person. Considering social exchange theory, we included shareable resources for the players to create a space for them to share resources and depend on each other.

Building on this social exchange tradition, Molm (2010, pp. 1–4) argues that reciprocity is not simply an outcome of exchange, but a structured process that varies across different forms of exchange. According to Molm (2010), reciprocity occurs when individuals provide benefits to others while being uncertain if they will receive those benefits in return. These forms of exchange create different social dynamics and significantly influence the emergence of trust and solidarity through mechanisms of risk and uncertainty.

Molm’s (2010) paper supports our study, since our game is built around resources, information exchange, and decision-making among the players. By creating these exchanges of information under uncertainty, the game prompts players to evaluate each other's intentions and future actions continuously. Thus, they must decide whether to share information, cooperate, or protect themselves. This constant decision-making creates moments for players to reflect on how trust is formed, maintained, negotiated, or challenged through interaction.

Mechanics such as can't reveal, prompting players to consider whether the co-operation is worth the risk and whether the rewards of trust can outweigh the risks of betrayal.

In this sense, trust becomes more than just a gameplay necessity, as the system creates interpersonal tension and conflict between cooperation and self-interest. Similar to Molm's (2010) discussion of reciprocity under conditions of risk and uncertainty, the players must navigate situations in which dependence on the other player can either strengthen cooperation or expose them to betrayal.

To establish trust-grounded mechanics, we need to understand how trust is modelled, the factors that constitute it, and how it can be assessed. Therefore, we decided to utilise Cho et al.'s (2015) work on modelling trust.

This paper provides a multidisciplinary survey of trust, reviewing definitions, related concepts, representations, properties and formulations of trust constructs, and identifying key factors that affect trust assessment. The authors propose that trust can be understood as a complex phenomenon rising from multilayered dimensions of Communication, Information, social and cognitive domains how these different components of trust can be mapped to different layers of trust attributes through the measurement of trust attributes from a complex, composite network consisting of communication, information, social, and cognitive domains, and examine how trust can be defined, measured and modelled across these domains. (Cho et al., 2015, pp. 13-14). Using this research, we analysed the factors across multidisciplinary domains that affect trust and used them as dimensions of trust, which we aimed to embed into game mechanics.

To successfully design mechanics that create space for reflection, it is important to understand how they affect dynamics and aesthetics. Therefore, we used the MDA framework as a design tool to understand and guide our design. The MDA (Mechanics, Dynamics and Aesthetics) Framework (Hunicke et al., 2004) is a formal approach to understanding games by helping designers analyse and refine the implementation of the game's result. This method helps clarify and strengthen the iterative process, making it easy to decompose, study, and design a broad range of game designs and game artefacts.

3. Game overview

In the Deep Games course, the original prototype was called *"Mines Away,"* and during the early stages of this thesis, it evolved into "*Act of Survival*". Before the design process started, we discussed our opinions on what trust is and how central it is in our everyday lives, and we based the early design choices on our own experiences with trust.

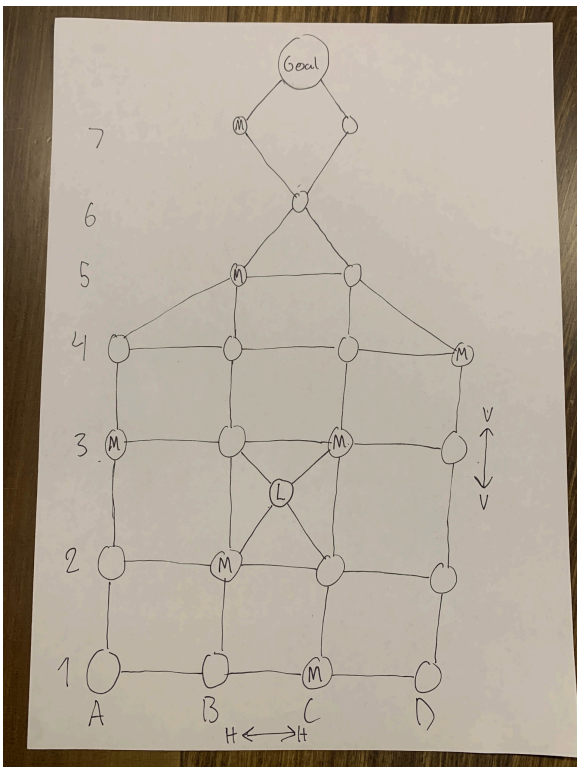


Figure 1: Work in progress on the game board. An overview of the initial design sketch of the game board, which shows the placement of mines.

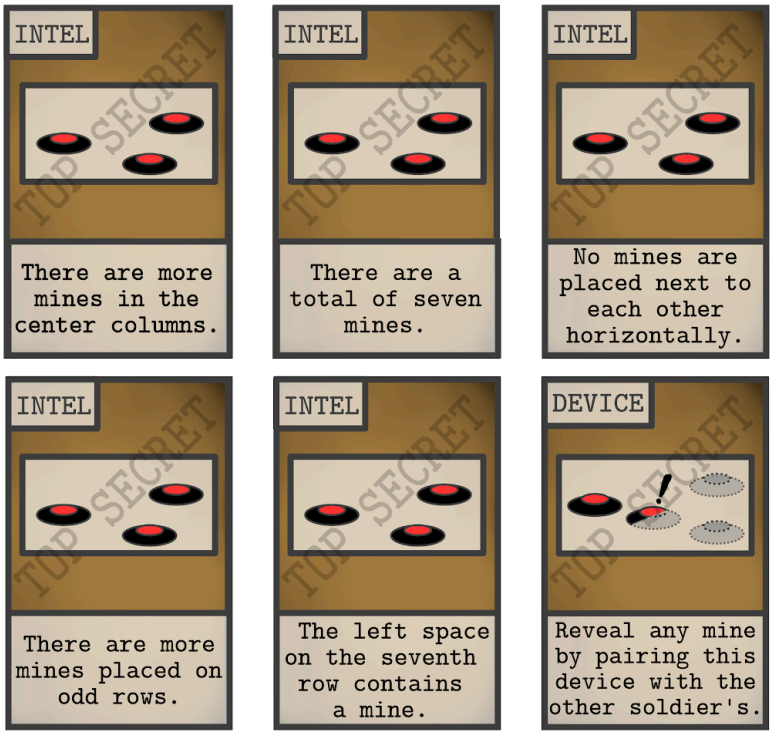


Figure 2: Overview of original clue cards and device card. The images show information from all five clue cards and a device card, which the intel player will possess. The intel has

five clue cards and draws one each turn, providing the intel player with information about the locations of the mines.

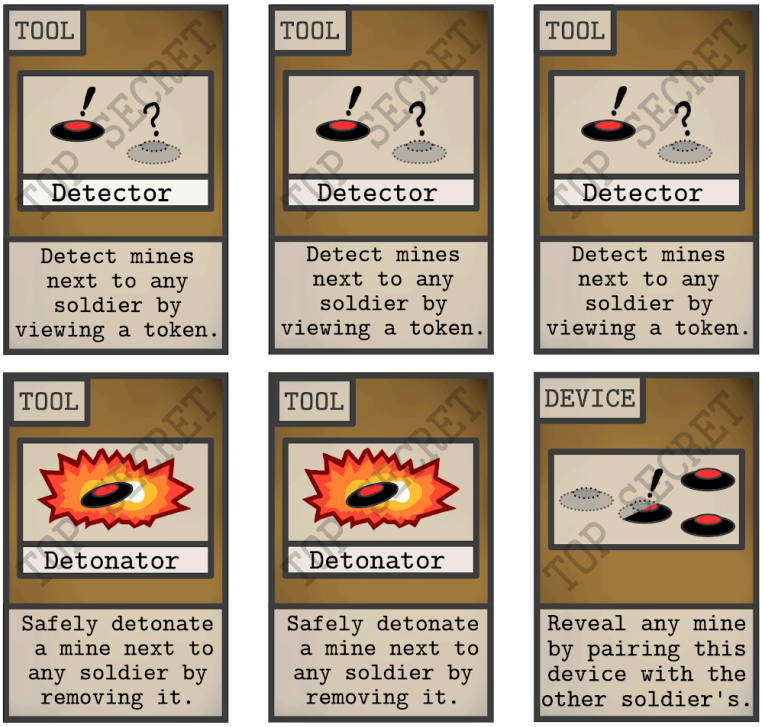


Figure 3: Overview of original tool cards and device card. The image shows information from all five tool cards and a device card that the tool player will possess. The tool player has access to three detectors, two detonators and one device card from the start.

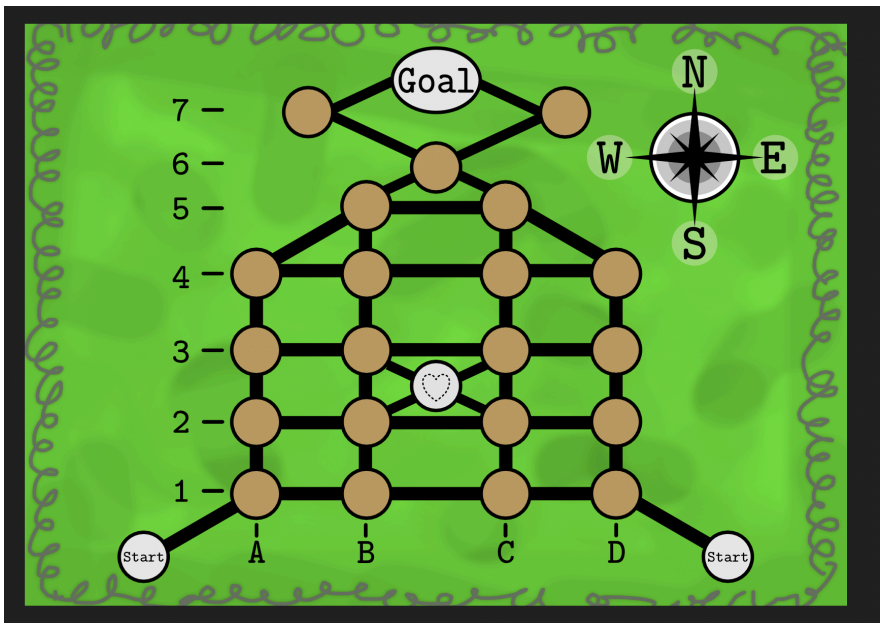


Figure 4: Overview of original game board. The image shows the original concept of turning a map into a game board for Mines Away. The map shows where the players will start, where the life token is, row numbers, the compass and where the goal is.

"Mines Away" was a two-player simultaneous turn-based strategy game, where each player is given the role of a soldier in a minefield. Each character possessed a unique set of abilities, with one having a device card and five clue cards as intel. Another with tools having two detonators, three detectors and a device card. The aim was to use war as a metaphor, let the players build trust in a vulnerable and treacherous environment, and have them experience what it takes to give and receive trust in a challenging situation. The game board represents a war minefield, with two player spawn points and 23 spaces to traverse: seven mines, 14 empty spaces, one space for an extra life token, and an end goal where players can get away from the mines and win.

Our goal for our thesis was to create an analog multiplayer game about trust, and we designed an environment where players' goal was to reach a safe zone together while starting from different locations. The players need to take risks while being vulnerable; each player has their own set of valuable tools/information that the other player will greatly benefit from. The goal here is to create a situation in which the players have the freedom to explore trust and its dimensions as they choose. When designing for trust, it is also important to design for potential betrayal, since without the possibility of betrayal, there cannot be trust.

The game *Act of Survival* is about two soldiers from different nations who are stranded on a mountain in a warzone. They are the only ones left, and they need to reach the top of the mountain where an extraction team from an unknown nation is waiting. The mountain is filled with landmines, and traversing this mountain alone is a hard task.

Each player is equipped with their unique abilities, one of which is receiving intel about the general locations of the mines. The other one has a select number of tools to either reveal a location or destroy a mine. Both players were equipped with a device card which can be paired up with the other soldier's device card in order to reveal a space on the map, meaning it's a shared resource. In the centre of the mountain, there is an extra life that one player can pick up and use for either of the players; however, it is surrounded by mines. The player can communicate freely but cannot reveal their abilities or information directly to the other player. The players must move simultaneously during each turn. To win, players must reach the safe zone either together or individually.

The exchange of individual resources is based on the Social exchange theory, which originates from Homans (1958, p. 606) proposal on social behaviour as a system of exchange, in which individuals trade, including both material and non-material goods, such as symbols of approval or status, where individuals seek to benefit from rewards that outweigh the possible costs. Our goal here is for each player to have valuable assets that the other will benefit from, and to foster a social exchange between them.

As our aim includes trust as a key component, we have utilised Cho et al. (2015) survey on trust modelling, and, based on our definition on the works of Gambetta (1988), Mayer (1995), Rotter (1980), who discuss major factors risk, vulnerability, dependency, uncertainty and possible betrayal that constitute trust. We want to create a game system centred on narration, rules, and game mechanics, in which these factors are embedded, enabling the establishment and exploration of trust.

With our system and mechanics, we wanted them to prompt the player to reflect on their decisions and experiences regarding trust. We did not want to give them a straight answer that trusting and helping each other is the correct way; we wanted to create a space where they could reflect on that on their own. Therefore, we base our work on Reflective Game Design made by Khaled (2018). It was important for us to give each player unique tools and shared resources, such as extra lives. Meaning that if they wish, they can cooperate to benefit them both, or go their own way.

Social Exchange Theory (Homans, 1958)	Unique and tradeable resources
Reciprocation (Molm, 2010)	Provide benefits under uncertainty
Trust modelling (Cho et al. 2015)	Understanding factors from multiple domains that affect trust
Trust Dimensions (Gambetta, 1988), (Cho et al. 2015) (Mayer, 1995) and (Rotter, 1980).	Implementation of the trust dimensions through the mechanics
Reflective Game Design (Khaled, 2018)	Creating spaces and opportunities for players to explore different possibilities - Player agency
MDA Framework (Hunicke et al., 2004)	Designing the mechanics

Table 1: List of theories

This list shows each theory and source utilised in developing the framework that guided our design and analysis.

Based on our theoretical framework and our design choices, we have a clear direction for our research. Based on the theoretical framework, we want to look into how resource sharing as a mechanic, shared benefits, uncertainty of the mines and the other players' intentions, traversing a treacherous and vulnerable environment together, affects players decision making and how they explore trust when they are placed in the situation with the trust dimensions (risk, vulnerability, dependency, uncertainty and potential betrayal) influencing them.

4. Methodology

4.1. Research Through Design

According to the authors Godin and Zahedi (2014, para. 2), who conducted a literature review, Research through design (RtD) is a research methodology that focuses on insights gained through the design process to understand a complex issue or question better. Research through design constructs research and new knowledge, meaning that researchers who use it are experimenting with new designs or products. (Godin & Zahedi, 2014, para. 1)

Research through design is part of the design process, where the main goal is to answer a research question, but the end product of the design is also important. The design should be evaluated, improved, and adapted throughout the process to create a better product that can ultimately lead to new knowledge regarding the research goal. Meaning that the researchers have to test their design, use it to collect data, identify flaws, and improve them. To use this framework successfully, the researchers must establish how they would gather their data, analyse it, and draw conclusions from it (Godin & Zahedi, 2014). We gathered qualitative data over three iterations through semi-structured interviews along with observations. We analysed the data using thematic analysis after each iteration, identified themes, and addressed gaps in our research question before the next playtest.

4.2. Data Collection and Analysis

4.2.1. Playtesting and Data Collection

We conducted playtests throughout the process to test our research question, *"How can we create a game system that creates a space for players to explore and reflect on their relationship with trust through the interactions with another player?"* Our goal was to conduct three playtests, each with two playthroughs, for a total of 11 participants. Between each playtest, we analysed the data, evaluated the design and made improvements before conducting the next playtest. We focused on qualitative data rather than quantitative data. By following this structure, our goal was to have a design that fulfills our research goal and addresses our research question by the end of the project.

For our research, we used a mixed-methods approach to qualitative data collection, combining semi-structured interviews with observations of the players while they played and interviews held afterwards.

We used semi-structured observations to capture player behaviour connected to the trust dimensions as they occurred naturally during gameplay. We conducted a non-participant observation, as we did not want to influence or disturb the participants while observing from a distance. We also conducted an overt observation, letting participants know we were observing them, as it would not be ethical in our situation to gather data without their knowledge.

Kumar (2023, pp. 2-3) highlights the importance of planning in order for the method to be effective and multiple questions need to be answered, including what should be observed, how the observation should be recorded, what Tools and techniques will be used, how to ensure that the observation will be accurate and what the relationship between the observers and the observed is. According to Kumar (2023, pp. 5-9), there are multiple ways to conduct the observations. It includes structured versus unstructured observation, participant versus non-participant observation, direct versus indirect observation, and covert versus overt observation. Through semi-structured observations, we were able to observe gameplay and identify the presence of dimensions, mechanics, and verbal reflections from players about trust and their interactions in real time.

For our research, we used semi-structured interviews to allow participants to reflect on their experiences, decisions, and thoughts about trust during the game, while also providing opportunities for follow-up questions and deeper inquiry. We conducted the interviews in accordance with the guidelines created by Tegan George (2023). According to the author, semi-structured interviews combine structured and unstructured interviews. These interviews mostly use open-ended questions, allowing the interviewer to be flexible and ask follow-up questions. With the help of semi-structured interviews, we were able to go deeper into the players' experiences and reflections around the different trust dimensions.

4.2.2. Data Analysis

To make sense of our qualitative data, we used thematic analysis to identify patterns in participants' responses. We used Lichtman's (2013) work to guide our analysis, which includes six steps. We used all six steps for the interviews; however, for the observations, we looked only for patterns related to the predefined dimensions and then directly created themes from them.

Step 1: Initial coding. Going from responses to summary ideas of the responses. The researchers should not interpret the data here; instead, they should use the participants' own wording to remain true to the data and develop descriptive codes. For every response, we assigned a descriptive code that summarised the player's actual words into short phrases.

Step 2: Revisiting initial coding. The researchers should now revisit the initial coding. During this step, we revisited and analysed our descriptive codes to find any previous mistakes and make improvements.

Step 3: Developing an initial list of categories. Based on the codes, the researchers should now create categories. We looked for descriptive codes with similar patterns and used pattern coding to create category labels that assign meaning to the codes.

Step 4: Modifying the initial list based on additional rereading. At this step, researchers should re-read the data, if necessary, and modify the list of categories. After we created the initial list of categories, we revisited the data to identify flaws and make the necessary adjustments.

Step 5: Revisiting your categories and sub-categories. During this step, we rechecked the categories to ensure they were accurate and, if needed, created new categories or subcategories.

Step 6: Moving from categories to concepts. At this step, researchers should develop concepts or themes from categories (Lichtman, 2013). Lastly, we identified similar categories and developed themes that were used in the thesis write-up.

4.3. Positionality

We are two master's students in game design with a fair amount of experience designing board games. We worked on the game *Act of Survival* before this thesis project, and we are both passionate about it and want to see it succeed. Both of us believe that trust is central to every relationship, whether romantic or professional, and it is worth exploring how our game can create reflections on trust. Without it, you cannot build a strong and true relationship.

We are aware that, because our views on trust may differ from others' and because of our desire to see our game succeed, we bring biases into this project. To avoid this, we aim to stay true to our data, background and theoretical framework to make grounded claims.

4.4. Ethical Considerations

For our research project, several important ethical considerations are worth keeping in mind. Before each playtest, we explained to the participants the purpose of the study, how we collect the data and how it would be used. The participants were anonymous to protect their identities and integrity. All data that could identify the participants was kept confidential, and their names were replaced with code names, for example, participant three.

All participants could choose to withdraw their data from the study at any time and leave during a playtest or interview. All data was kept on a personal hard drive, regardless of whether it was transcripts, recordings or other data. This information was given to the participants before each playtest, and they were required to sign a document granting us consent to use their data.

4.5. Limitations

In this section, we will discuss and present the major limitations of our study, how they may have influenced the results, and potential future research directions.

The first major limitation is that all participants were game design students, which limits the results to their opinions. To obtain more accurate results, additional research could be conducted with a greater variety of participants.

The second major limitation is the low sample size: in total, we had 11 participants, of whom one played both iterations one and two. Future research could conduct a similar study with a larger number of participants to obtain more data for analysis.

The third limitation is that the qualitative data were interpreted by only two researchers, which may have introduced bias and subjectivity in how the themes were identified. Future research could be conducted in a larger team, with more researchers interpreting the data to reduce bias and subjectivity.

5. Results and Analysis

This section discusses each prototype and iteration, as well as the results from all playtest sessions. There were six playtests conducted across three playtest sessions. All the playtests were conducted in a university-provided student room without external disturbances. In total, there were 11 players, with each playtest involving two players of mixed genders and ages, all of whom had backgrounds in game design or creative fields. We recruited the participants through the university, mainly classmates from earlier courses. The relationships between the participants varied, from a couple to total strangers. Rather than focusing on how their relationships affected the outcome, we examined when players made trust decisions and when reflections happened.

5.1. Iteration 1 of *Act of Survival*: First Prototype and Initial Playtest Session

5.1.1. Prototype 1

Building on the mechanics of the prototype developed in the previous course on deep games, the *"Act of Survival"* prototype was enhanced with a new narrative and the appropriate art style for its components. It primarily focuses on creating an environment where players can explore trust in their own way, aligning with reflective game design grounded in social exchange theory. Thus, the prototype aims to connect mechanics to theory, creating gameplay that can represent trust elements.

It follows the storyline of soldiers trapped in a minefield; they are from different nations that are enemies in terms of international relations. This allows the players to take different characters with unique abilities and resources. By having the players enter the game board from different sides, and by having them be soldiers from different nations, the goal was to create a sense of uncertainty, suspicion, and difference among the players. However, through dependency, cooperation and a shared goal, we aimed to encourage the players to move towards the helicopter for collective safety.

The military theme was chosen because soldiers are constantly put in dangerous, vulnerable situations where they have to trust their fellow soldiers to have their backs. However, we wanted to go deeper, challenge trust further, and gain more insights into how trust works in

games in real time. Therefore, we placed the players in an uncertain situation in which they were on opposite sides.



Figure 5: Game overview of the first iteration of Act of Survival. The image shows the complete first iteration, including the new map, rules, intel cards, tool cards, and each player's introduction cards, which explain their roles and abilities.



Figure 6: Overview of playing cards in Act of Survival. These cards offer different mechanical options to each player: the intel player receives information about the location of mines, and the tool player receives tools to detect or neutralise them.

As shown in Figure 1, the game includes a game board, an instruction manual, two player tokens, and player resources. The game board is visually designed to represent a path of mines on the hilltop, containing two player spawn points and 23 spaces to traverse, including a path of seven mines, 14 empty spaces, a space for the extra life token, and an end goal: a rescue helicopter. The instruction manual helps players read through the narrative and understand the game's rules. Each player has two sets of individual resources and a common resource called extra life, which both can use.

Starting with the resources of Player A, they have five cards stacked in order, named as intel, and have information about mine locations in an indirect way that they are receiving from the headquarters of their character's nation. At the same time, Player B has two distinct tools: detectors and detonators. There are three detectors and two detonators represented through physical tokens and cards. Each player has information and usage about their resources on separate information cards. The players are required to reach the goal (helicopter) by

traversing spaces while avoiding mines, using their resources and exploring trust in their own way.

By giving players varied abilities, we aimed to give them their own strengths and weaknesses, prompting them to depend on each other through their skills while also making them vulnerable to each other by not being able to protect themselves fully. By dividing player roles with tools and intel, the game encourages them to communicate and share their resources. With intel, one player has information about the locations of the mines; while holding tools, another player can detect and destroy mines, and together the two players can communicate, strategise, or go their own way.

5.1.2. Design choices prototype 1

As Khaled (2018) states, reflective game design is about allowing players to explore different possibilities and draw their own conclusions rather than giving them a correct way to play. Therefore, our goal was to design the system so that the players had the option to cooperate, deceive, or betray, and we aimed to have each dimension present throughout the system and its mechanics.

Simultaneous move

Players move simultaneously each round; this mechanic was designed to introduce trust dimensions, uncertainty, and risk among the players. Through this mechanic, we tried to encourage players to anticipate the other player's movement as they considered their next move, and if they agreed on where to move, they had to trust the other player's word.

Can't Reveal

Players are restricted from showing their information directly, but can speak or share in any desired way. Uncertainty, potential betrayal and vulnerability were the targeted dimensions through this mechanic. Even when a player tells the truth, with information hidden, the other player can never fully know whether they are telling the truth, leaving them vulnerable to betrayal and deception.

Detonators, Detectors and Intel

These are the resources distributed uniquely among the players. By providing different resources to players, our goal was to make them tradable and encourage sharing among them. By making them unique, the system was designed to create a dependency between the players, since reaching the rescue helicopter without the other player's resources would be difficult.

Extra Life and Devices

These are shared resources, extra life is claimed by one player and can only be used by any of them once, whereas devices are distributed among both players and need to be used together.

These mechanics are implemented to support player decision-making and give them a choice about whether to cooperate.

All the mechanics remained the same as in the initial idea, whereas visual changes to the map and narrative changes were made.

Map Design

During the first iteration, the map was completely redesigned, and the minefield was replaced with a forest-covered mountain cliff theme. The decision regarding the new design initially felt intuitive, but upon reflection, it revealed that the mountain evoked a strong sense of risk and danger. Unlike the minefield, which was empty and could be perceived as something players might avoid, the mountain is covered with trees and visual obstacles, with the spawn point below and the finishing point at a cliff's edge, which becomes a central obstacle that players have to face.

Narrative

The helicopter, with a mountain setting and backstory, was the narrative change implemented in the first iteration. The first thought about the game is how and where it is going to end; having something at the goal is about immersing and investing players in reaching the end. Especially replacing minefields with mountain themes, we wanted to create a believable survival scenario rather than something one could walk away from. The mountain replaced the empty minefield with lush forest, rocks, dirt and a rough path that does not affect the game mechanics but is intended to create a strong sense of fiction with risk and danger. By having players start in the shoes of soldiers from different nations and sides of the map, but with a shared goal of reaching the helicopter, we aimed to create space for both conflict and collaboration.

5.1.3. Playtest Session 1

With the required information and assets in place, the first playtest focused on gathering insights into trust dimensions during the gameplay, with a focus on reflective game design. To obtain the data, semi-structured observations and interviews were conducted. As of Playtest Session one, four players were involved, spread across two playtests.

5.1.4. Observations

During this phase, players have been observed for signs of dimensions through verbal communication and actions. We looked for moments when the players reflected verbally or made decisions regarding specific trust dimensions, such as dependency, potential betrayal, and risk, as well as trust as a general concept. The collected data were structured as themes.

Main themes that were present in the observation data

1. Dependency between the players

“In the beginning, I said we both need to be safe, right?”

After using many tools, the tool player said they sacrificed too many of their tools for the intel player and further said they were considering going alone. Then the intel player reminded the tool player that they wanted both of them to be safe from the beginning and were trying to reveal information accordingly. This further says how the intel player was reflecting on how dependent they were on the tool player.

Overall dependency between the players was observed to be high throughout both playthroughs. In one of the playtests, when the intel player collected the extra life token, the tool player noted that the extra life tool has huge potential to revive a player fully and reflected on the benefits of cooperating and being dependent on the other player. Besides that, they relied on each other and each individual's abilities in order to safely traverse the minefield. Except for the earliest steps, players often tried to plan where to go, were sincere in sharing their intel cards, and used the tools to help reveal or destroy potential mine locations based on the intel to move safely. There were some moments when the players took steps without relying on each other, but that was rare. As a result, all four players survived and completed the game together.

Unique Resources and the fact that they were tradeable/shared among the participants worked well, creating dependency and cooperation, which is supported by social exchange theory. Individual resources are clearly connected within social exchange theory (Homans, 1958), in which players' physical resources serve as material goods. In contrast, information about the resources they hold serves as a non-material, symbolic good that they can exchange.

2. Players took risks

Throughout both playthroughs, the players took many risks by trusting each other's information and advice, relying on their individual abilities and information. A common pattern in both playthroughs was that the tool player took a step on the board based on information provided by the intel player; they did so even though they did not know whether the intel player was telling the truth. There were multiple times when the tool player questioned whether the intel player was telling the truth; however, they still took risks.

“It is both interesting and hard to make trust-based decisions when you do not know the other players' cards.”

In this quote, the tool player expressed the difficulty faced and the risks taken in trusting the information shared only through the intel player’s words.

Unable to see other players' information directly and relying on separate resources, the players took risks based on other players' words and promises. For example, if the intel player went to a certain location, the tool player would help the next round by using either detect or detonator to help the intel player.

3. Players showed uncertainty around the location of mines and each other

“I do not have to work with you; I can go alone rather than depending on you.”

“How can I be sure you are right?”

Both players showed uncertainty around the location of the mines and each other's true intentions in the game. Especially in both quotes above, the tool player mentioned on multiple occasions that they were unsure whether the intel player was truthful and whether the player actually tried to help rather than using them to their own benefit.

The players were vocal about their concerns regarding the location of the mines and expressed uncertainty about where the safe path was and how to traverse the map safely. In the first playthrough, there was more doubt and uncertainty about the players' intentions, and less in the second.

“I do not know how powerful your tools are. Can I trust you to help me in this situation?”

In the quote above, the intel player expressed uncertainty about trusting the tool player because they did not reveal their tools and did not know whether the tool player would use them to help. Not being able to share their information directly (can’t reveal) created uncertainty among the players about their intentions, resources, and whether they were telling the truth. This mechanic made potential betrayal possible; the intel player chose not to share a piece of information and even lied at one point.

This mechanism developed from social exchange theory and the trust-modelling process described by Cho et al. (2015). Players cannot reveal their unique resources directly. As explained above, information from unique resources serves as symbolic goods that they can exchange among themselves. This way of sharing information clearly raised doubts among players about their opponent's honesty in sharing information.

4. Lack of reasons to distrust/betray

During the playthroughs, there were barely any signs of the players considering betraying or distrusting each other, since there was not enough reason or incentive to betray. There were a few reflections on whether one player could trust the other, but mechanically there are not enough benefits or reasons to go down this path.

The players reflected verbally on multiple trust dimensions, such as dependency, risk, and potential betrayal. The players asked both themselves and the other player questions regarding their decisions based on trust, which is crucial for reflective game design (Khaled, 2018). There was, however, not enough conflict among the players or reasons to distrust, creating situations in which players realised that cooperation was the only viable way to win the game, which runs counter to reflective game design. This needs to be improved in future iterations to make the reflective game design stronger in our game.

5.1.5. Interview Results

After the playtest, players were interviewed semi-structurally, and the data from both interviews was analysed using thematic analysis, resulting in five major themes that discuss the highlighted dimensions and their connections to observations.

Sharing resources and dependency (14 instances)

The players stated that by collaborating and sharing resources, one can also receive and rely on help from others. During the playtest, dependency among players was high; they shared resources, helped each other, and ultimately won the game. In several instances, they mentioned that they were trying to be supportive and that, after receiving help, they wanted to help other players in return. This shows the exchange of resources between them, connecting to our base theory of social exchange, depicting the transaction of goods between people, with rewards and costs. Moreover, the continuous reciprocation of these exchanges highlights how they want to balance the rewards they receive against the costs.

“I had some tools, but they didn't feel like enough to achieve it on my own. I later realised there were three detectors, so if I start here, it will still feel difficult. So I realised it later.”

The tool player mentioned that they tried to strategise during gameplay. They said that they initially thought they could traverse the minefield and achieve on their own. Only after counting their resources and checking the pathway did they realise it could be difficult to do alone, which suggested they choose to depend on the intel player and share resources between them.

During the game, players took risks since they had chosen to trust the other players’ word about their information/tools. Though it shows these risks are validated through the gameplay, as they received benefits in taking risks and reached the goal.

Potential deception (11 instances)

During the playtests, players mentioned on multiple occasions that they could either try to deceive each other or not collaborate. They mentioned how powerful game elements are, saying that their tools and the can’t reveal mechanic gave them opportunities to deceive. Players further added that each other's actions showed how careful they were, and that this also raised doubts about being deceived, but the information sharing made it possible to take risks and rely on each other. Even though the players reflected on the possibility, the majority chose not to pursue deception.

“I find it quite interesting. That is, it is a game where you can be as honest as you want with the other person. Like you do not have to share or help at all.”

A tool player highlights different possibilities they have while traversing the minefield and exploring trust, further adding that if they do not want to, they do not have to share or help the intel player at all, stating co-operation is a choice rather than fixed.

“If I think I cannot believe him, I should be able to protect myself. But I felt our relationship is fine, and I can give him another chance.”

An intel player said they were doubting the tool player and could not understand the tool player’s actions and intentions. However, they felt that, in that situation, their intentions towards each other were fine and thought the tool player could be given another chance to understand better.

On multiple occasions, players tested one another to determine whether the other was telling the truth and to protect themselves. They said they were trying to follow different strategies and tried to observe facial and vocal cues. Even though they recognised the risks and potential for deception, they still tried to give each other another chance to understand each other's attitudes towards trust.

Not enough reasons to distrust or betray (9 instances)

Players stated that the stakes were not high enough to betray their opponent. They said their collaboration and lack of reasons kept them from thinking about competing, betraying, or worrying about potential betrayal. Trust gets tested under pressure and in challenging situations, and, aside from the dangers of the mines, there was not enough pressure or vulnerability for the players to experience potential betrayal. They further suggested that competitive elements could increase the stakes and give players more choices.

“We collaborated well, which reduced the feeling of betrayal, and we don’t need to worry about that. Without collaborating well, we have to be concerned about trust and betrayal.”

A tool player expressed that the game is about trust and betrayal, people who either have information beforehand or not would still say it in the same way. They further say that before the game, players need to believe in trust, which basically worked in their duo, resulting in good collaboration and reduced feelings of betrayal; otherwise, there will be worry about trust and betrayal.

“So, if there were more competition in some way, then it would make it less reason to trust.”

A tool player said there were few reasons to develop distrust between the two players, and it felt like the more beneficial strategy was to collaborate. They added that, if the stakes were higher, that could create more friction or competition between them.

Uncertainty towards the other player (7 instances)

Players expressed doubts about each other’s actions, as they had limited information about each other's abilities and intentions. On multiple occasions, the players expressed uncertainty about their true intentions and explained that they wanted to protect themselves. Even though they showed uncertainty towards each other, the players took risks as they listened to one another's information and suggestions. So, they had to take a risk in believing what the other player said.

“You mentioned you cannot collaborate with me. I think it is understandable. If I find this guy I cant believe him, then I should be able to protect myself. It is just not very serious, that sentence, that word, that speech”

An intel player expressed doubt about the tool player’s intentions as the tool player received information. By analysing the verbal cues, they said they could not believe the information shared about the tools was true and that they might need to protect themselves.

Agency to choose your own path (6 instances)

Players mentioned that the game provided them with the freedom to choose their own playstyle and to explore the game's strategy while playing. Specifically, the "Can’t reveal" mechanic gave them an option to share the information in their desired way, either collaboratively or non-collaboratively. Players expressed curiosity about testing the gameplay in a non-collaborative way, but instead chose to collaborate and share information in a non-competitive way.

“I am also curious, like, how it might work if we did not collaborate and try to just do our own thing.”

One of the Tool players said they liked the gameplay experience because both players shared information and collaborated. However, they were also curious about what would happen if they did not collaborate and went down their own paths.

5.1.6. Findings of Prototype 1

Main insights and shortcomings

How the design created conditions for players to engage in trust reflection

People reflected on multiple aspects of trust dimensions during gameplay, such as dependency and risk. Social Exchange Theory has been shown to increase dependency and cooperation among players. From the observations and interview data, it came to the surface that prototype one achieved in making players realise about trust dimensions, risk, uncertainty, dependency and betrayal, but how their occurrence in the game varies at different levels.

The Unique resources and can’t reveal mechanic played a crucial role in paving the way for multiple trust dimensions to emerge, thereby making it possible for trust elements to emerge. Having their own unique resources and relying on other players’ resources, not by revealing but solely through communication, created uncertainty about how much information is true. This led to the risk of proceeding with uncertain information and the dangerous surroundings in which the players were.

The can’t-reveal mechanic promoted uncertainty directly about the other players' true intentions. Since the players could only talk about their resources rather than show them

directly, they questioned each other's honesty, which led to uncertainty about their intentions and goals. This was reinforced on multiple occasions when players were hesitant about revealing their intel. This mechanic also created reflections on deception; players mentioned that it made it possible to go down that path, opening the door to potential betrayal.

Players considered the extra-life mechanic a powerful tool that enabled player dependency. An intel player who picked up the extra life token said they felt powerful in holding the opportunity to help and share the tool. In contrast, the tool player said they considered the player holding the extra life token powerful, and it was not certain whether the person would decide to help or not, which made one dependent on them. This shows player behaviour around another player, focusing on mechanics, and displays different dimensions of dependency and uncertainty among players. The player with uncertainty also shows traces of vulnerability in how they reflected on the other player's power over them.

What does the current prototype lack to create?

The mechanics in the current prototype failed to create sufficient vulnerability and potential betrayal between the players. Two of the major reasons are that the stakes were not high enough or that the situation did not create enough conflict between the players; therefore, there were not enough reasons, benefits, or incentives to betray the other player. Although players were given agency to choose their own path, the lack of competitive elements and higher stakes reduced the potential for deception and betrayal, leaving players to lean more towards collaboration. For trust to fully emerge, these trust dimensions need to be more prominent in the game.

What can be iterated for the next prototype?

Based on the above findings, we aim to improve the presence of vulnerability and betrayal in the next iteration. The fact that both players can survive until they win, without affecting the other player, makes it non-competitive and lacks high enough stakes for the players to reach the goal. Along with the lack of stakes between the players, there was also a lack of conflict. Our goal for the next iteration is to focus on the win conditions, increase the stakes and create a more competitive environment for players. Therefore, to model and improve the lacking dimensions, there need to be more elements in the game that set up stakes and create a direct conflict between the players.

5.2. Iteration 2 of *Act of Survival*: Second Prototype and Second Playtest Session

5.2.1. Prototype 2

Based on the findings from the previous prototype, several changes have been made to the game, including updates to the visuals and in-game narrative and mechanics.

5.2.2. Design choices prototype 2

In-Game

Starting with the narrative, removing neutrality, player characters were described as soldiers from enemy nations, which could create conflict from the beginning. Moreover, the story has been introduced with more fiction, written in a way that connects to reality so players can relate to it more, immerse themselves, and imagine it in their own ways. Enhanced fiction includes a signal jammer mechanic, a direct replacement for the extra life mechanic that follows the narrative and connects with real-world illustration. When it comes to the win condition, the rule for reaching the rescue helicopter remains the same: it is neutral to both soldiers, and whoever reaches it first will win, or they could both win by reaching the helicopter at the same time. This serves to increase competitiveness in the game. Lastly, device mechanic cards were renamed to old detectors with improved narrative, which also falls under the enhancement of fiction.

Visual changes

Narrative and Rules were separated onto two cards, creating greater distinction between them when reading. Players mentioned there were many elements on a small size board, trees were blocking the path, and the rows were not being displayed, making it difficult to find the spaces to traverse on the game board. To improve visual clarity, the game board was doubled in size, the trees in the game art were distributed, the path was darkened, and row numbers were added to make the path easier for players to track.

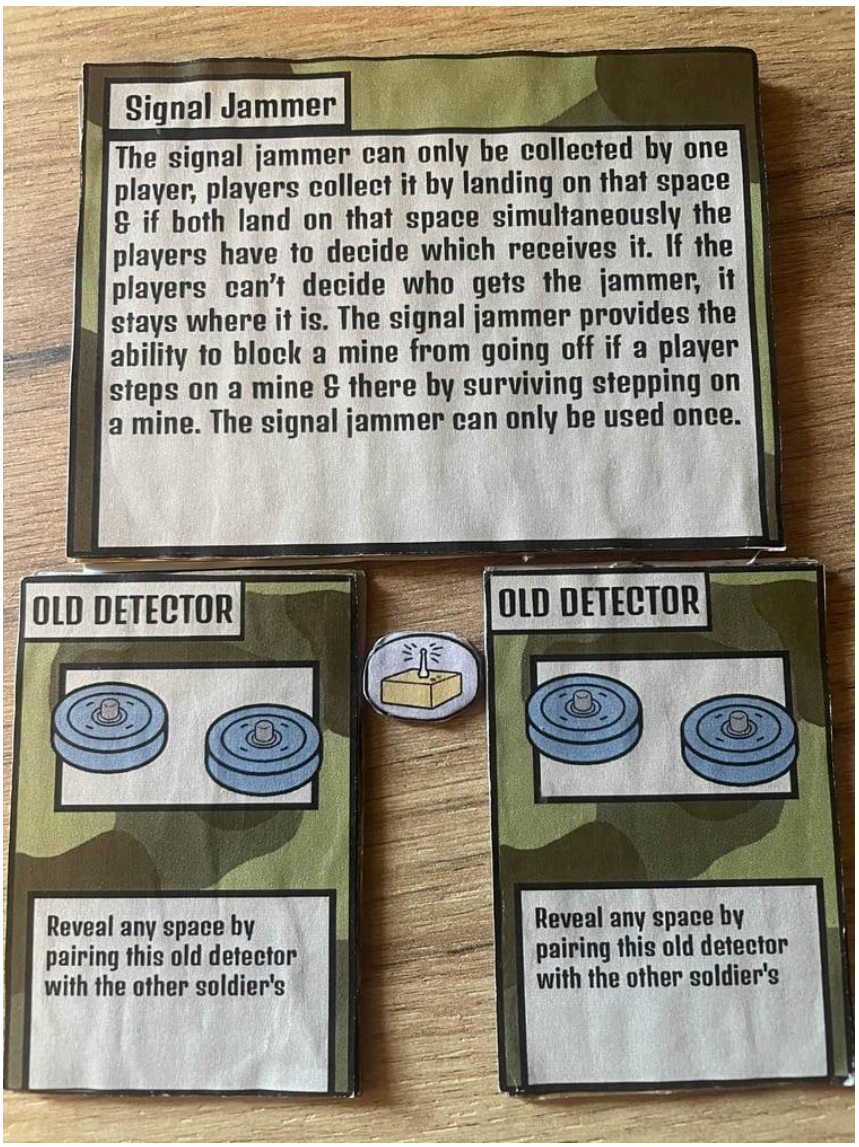


Figure 7: Overview of the new old detectors, signal jammer card and signal jammer token. The image shows two old detectors and the signal jammer cards; each player possesses one old detector, which they can pair to reveal a space on the board. Any player can pick up the signal jammer, and if a player steps on a mine, the player with the signal jammer can save the other player or themselves.

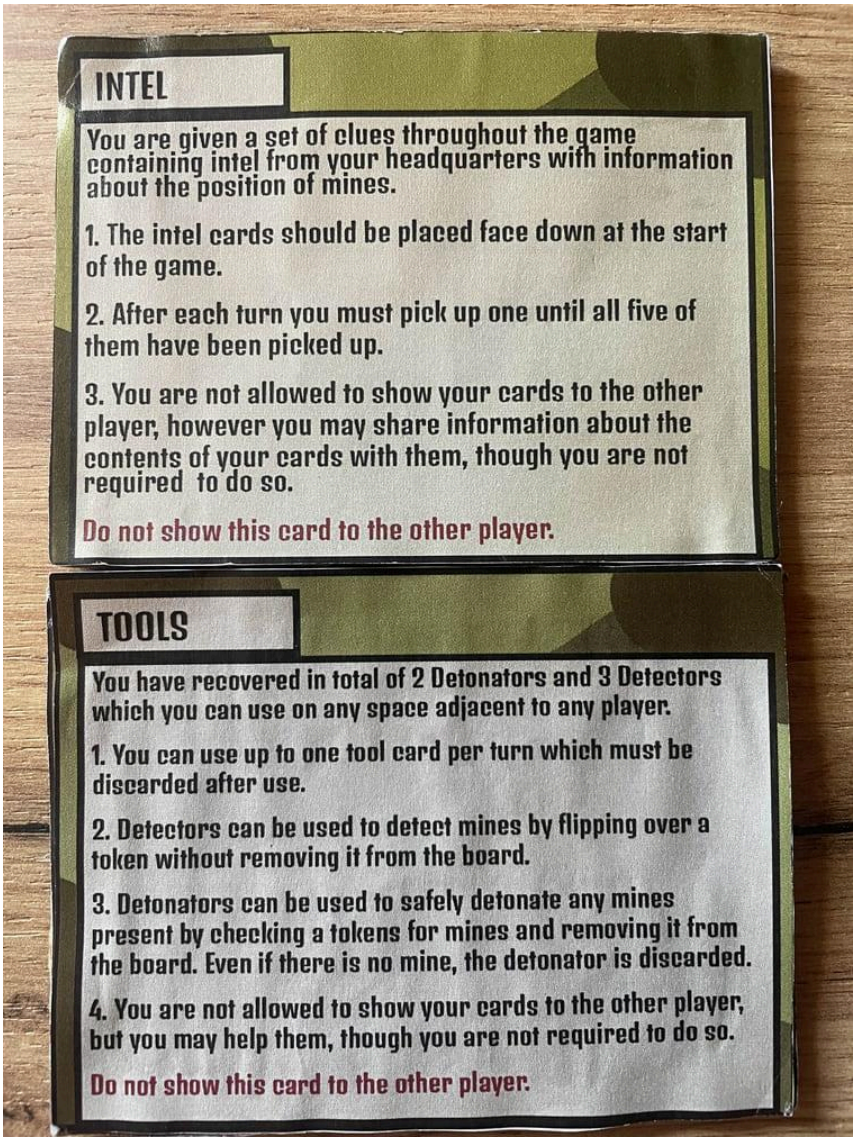


Figure 8: New player role information cards. The images show the new player role information cards with improved simplified sentences, the addition of numbers for each rule and new art.



Figure 9: New game board. The size of the game board was increased, trees were removed, and the paths were made darker to improve visibility. The board went from an A4 to an A3.

5.2.3. Playtest Session 2

Continuing from playtest session one, the second playtest session focuses on gathering data using a similar approach to gather insights into dimensions that emerged in gameplay, with a focus on reflective game design. To obtain the data, semi-structured observations and interviews were conducted.

5.2.4. Observations

Following a similar approach to the previous session, in this phase players have been observed for signs of dimensions through verbal communication and actions, and the collected data have been structured as follows. As of Playtest Session two, four players were involved, spread across two playtests.

Main themes that were present in the observation data

1. Uncertainty towards the other player's intentions

“Can I ask you what your information said? It’s hard to plan when I don’t know.”

During this playtest, the players showed more uncertainty towards the other player's true intentions. They did not reveal everything right away, and one intel player never revealed any information directly to the tool player. This made the tool player uncertain about the intel player's intentions, as they planned their next move and which tools to utilise. They questioned the intel player directly to get clear information on the mines' location.

The players did think and plan more before taking any action. The players were generally more cautious before they trusted the other player's information. When they noticed the other was hesitant to reveal information, they questioned each other more directly.

Because they could not directly reveal information about their abilities, the players left one another uncertain about their abilities and true intentions. During this iteration, players were more cautious about revealing their intel/tools, and there was a mix of players who completely revealed and those who kept it to themselves. One common pattern that differed this time was that players took longer to reveal or did not reveal at all.

2. Potential of betrayal

As players were more cautious, the new win conditions made the potential betrayal more prominent than in the previous iteration. Since some players did not reveal their information immediately, the other player became more cautious and aware that betrayal was possible. One intel player tried to use the tool player to test certain locations by telling them to go in a certain direction based on their intel, even though they did not reveal all of it. Another tool player said it was hard to decide where to go because the other player did not reveal their intel.

"This intel is so dangerous, I don't know what to do with it. We are enemies, it's hard to decide what to do."

At the end of the game, the intel player had information about the last mine but chose not to tell the tool player to protect themselves and potentially win. The intel player reflected on how hard it was to decide what to do with the last piece of intel.

Changing the win condition to the rescue helicopter taking off as soon as one player reaches it, or both players reaching it simultaneously, raised the stakes and made players more vulnerable to betrayal. The potential for betrayal was observed significantly higher than in the first iteration. Even if the players worked together earlier in the game, there was always a possibility that one player would be betrayed in the end.

3. Dependency between the players

Dependency between the players was consistent, but the presence was slightly lower than in the first iteration. This time, players were more hesitant to reveal their information, questioned each other more, and the threat of betrayal was more prominent.

“We should find a way to the endpoint together. I know you have some kind of tools; we can use our resources together.”

“I think it's still possible for us to survive even though we don't have many tools yet if we work together.”

In both of the above quotes, the Intel player expressed hesitation yet tried to rely on both players' tools and information to traverse the mountain safely. Even though, at some point, the players did not have many resources left, the intel player still expressed their dependence on the tool player to survive. Still, there was more potential for betrayal and uncertainty about the other player in this iteration. By giving the players Unique Resources and allowing them to trade/share them, a dependency was created among the players. As they relied on each other's tools and intel to survive, all players ultimately survived and won the game together.

“If you go and pick up the jammer since you’re ahead, then we both can take/use it.”

Even though the signal jammer was a direct replacement for the extra-life mechanic, the fictional element prompted more reflection and discussion in this iteration. In the quote above, the tool player discussed the potential of the signal jammer and how both players could use it to their benefit.

4. Risk-taking based on limited information

“I need to take a risk based on the limited information.”

“It is very dangerous to move on the mine field with little information.”

In both playtests, the tool players voiced concerns about intel players' abilities and the dangers of traversing the field with limited information, yet continued to take risks based on the intel players' words and the tools they had, in a way similar to the first iteration. However, in this playtest, players took more risks with less information, given the greater secrecy and caution about revealing their abilities.

The can’t reveal mechanic had players taking risks based on the words or promises made by each other, while never truly knowing the intentions. During the playtests, players discussed how to proceed and who should take which risks. By moving simultaneously, players take risks every round because they do not know whether the other player will keep their word.

5. Not enough vulnerability directly between/towards each other

Even though vulnerability was raised in this iteration, it still lags behind the other trust dimensions. Since there was a constant threat that the players could betray each other in the end regarding the win condition, we saw greater vulnerability and uncertainty between the players, and a more cautious use of the tools by the tool player. However, there is room for improvement in direct vulnerability between players, meaning the game mechanics could

make players more vulnerable to one another, rather than relying solely on the win condition or abilities.

The players reflected on multiple trust dimensions verbally during the game, including risk, uncertainty, and dependency. The players were more aware of betrayal in this iteration, but because they did not want to reveal it to the other player during the game, there was little verbal reflection on this dimension. However, they reflected on betrayal in the interview after the game. For this iteration, we improved how we modelled potential betrayal by changing the win condition, which proved successful because the potential for betrayal was more prominent. Dependency, uncertainty and risk were modelled similarly to the first iteration. To model all dimensions of trust, we still need to model players' vulnerability more effectively.

5.2.5. Interview Results

Following a similar approach to previous interviews, after the playtest players were interviewed semi-structurally, and, using thematic analysis, the data from both interviews was organised into six major themes that discuss the highlighted dimensions connecting to the observations.

Potential betrayal and testing trust (15 Instances)

Compared with the previous playtest, players' responses regarding the potential of betrayal have improved significantly in this iteration, as they were given more reason to choose to betray. Even though players have not gone down the path of betrayal, they mentioned that they thought about doing so while calculating the risks and benefits.

"For me, I think maybe at this point, he has enough resources to solve all this and to reveal this mine, and he can go and escape alone."

Players talked more about the new mechanic regarding the helicopter, which acted as an improved win condition. An intel player highlights the opportunity to betray the tool player at the very end and says it was clearly visible. The intel player also added that one can expect the same from the tool player, as they were concerned about the chances of going alone in the end.

"When he trusts me, I trust him, then I show that at that time, this place is very safe."

An intel player mentioned they are continuously testing the tool player's reliability and how much they can be trusted; this is one of the main reasons they are not willing to betray or collaborate on taking risks.

Sharing resources and dependency (22 Instances)

The current iteration led to greater sharing and dependency among players. Dependency occurred in 12 instances, showing that players relied on one another to share information and tools.

“I just tried my strategy on giving straightforward information, and luckily, we worked together in the end.”

An intel player mentioned that they followed the strategic information-sharing approach straightforwardly and that it luckily worked for them towards the end, as they all reached the helicopter together.

Deception (14 Instances)

Compared with the previous iteration, we have seen a rise in the potential for deception. Players were more hesitant to reveal their information and often revealed only certain parts of it. The players reflected more on whether to lie and how much information to share.

“I think for this role, there is plenty of possibility to lie to the other one.”

An intel player mentioned the different abilities each player role can have. They said that, as a player with intel, they have more opportunities to lie about the information while sharing it. They have also added that it is quite risky and it puts pressure on them to do so.

Trust reflections (8 Instances)

During this playtest, we saw participants reflect on trust as a general concept rather than as something directly connected to their experience of playing our game on multiple occasions. However, they mentioned that, in reality, people can offer each other in order to gain trust and benefit themselves, which shows their thoughts align with social exchange theory.

“I mean, during the playing procedure, it is very clear that we listen to each other and both of us can offer something and trust each other.”

A tool player compared the sharing of resources among players in gameplay to the real world. During gameplay, both players listened to each other, shared resources, and were dependent on each other. The tool player noted that the transaction works in the real world similarly, a point further supported by the intel player.

Agency to choose their own path (7 Instances)

Similar to the last playtest, players highlighted opportunities to go their own way in how they played. The participants discussed how they can choose whether to deceive, collaborate, betray one another, or win on their own. This clearly shows the improvement of paths towards betrayal and deception in the current iteration.

“So it can create more interesting scenarios based on how we play. We collaborated, but if you didn’t, maybe you could try to trick anyone, for example”

One of the tool players says that not revealing the information directly makes the gameplay more interesting. This gives players more agency to think for themselves and decide whether to collaborate or trick the other player, which creates more varied gameplay scenarios.

Vulnerable to mechanics (6 Instances)

During this playtest, players felt more vulnerable than in the previous iteration. Primarily, they felt vulnerable to the mechanics, such as limited information in the early stages and a smaller chance of survival if they did not possess the signal jammer tool.

“The problem for me is that, because I do not have any tools to solve mines, or to know where they are. So, that's why we had a concern on using a lot of tools in the beginning. Right? So, I was very worried about dying very early in the game .”

An intel player said they faced a problem without knowing the exact location of the mines or the tool player's abilities. They said they only had clue cards, and without any tools, they could not find the mines themselves and had to rely on tool players for tools. So, with limited information shared between the two players, the intel player, along with the tool player, had to use more tools than necessary.

However, there were limited reflections on being directly vulnerable to the other players and their actions. There was not enough that the players could do to each other to influence the other player's outcome, aside from deceiving and potentially betraying.

5.2.6. Findings of Prototype 2

Main insights and shortcomings

How the design created conditions for players to engage in trust reflection

Overall, the game successfully modelled agency for the players and continued to show similar consistency, with seven instances during the current playtest. The players also reflected on the dimensions in both the observations and the interviews. The biggest improvement since the first iteration is that the potential for betrayal is stronger through the changed win condition, comparing with interview data of the previous iteration, deception was increased by three instances, totalling 14 and potential betrayal increased from none to 15 instances, which is important for us to model all trust dimensions fully and in the end, have a game where players can reflect on trust. The unique and tradeable resources still modelled dependency whereas can’t reveal, and simultaneous move mechanics modelled uncertainty and risk.

What does the current prototype lack to create?

The main shortcoming was the players’ lack of vulnerability towards one another. Even though we still saw an improvement with the new win conditions, it was low compared with the other dimensions, so it was still not enough. There was no clear threat between the players, and aside from lying and deceiving, there was not much else they could do to each other.

What can be iterated for the next prototype?

With several changes made to the current prototype, it was still not a fully explorable system with all dimensions included. As mentioned above, with huge improvements in potential betrayal balanced by uncertainty and dependency, the game requires improvements in the presence of vulnerability between players.

The fact that players could betray each other, regardless of what happened before, introduces some vulnerability. Besides that, the players could not affect the other's outcome enough through the mechanics. For the next iteration, we focused on adding a new mechanic that puts players in a situation where they are directly vulnerable towards other players.

5.3. Iteration 3 of *Act of Survival*: Third Prototype and Final Playtest Session

5.3.1. Prototype 3

The third iteration mainly focused on implementing mechanics to improve risk and vulnerability among players. The major update here is the implementation of a new mechanic called ‘Trap’ and an extra tool, ‘Thermal Scanner’. Without altering the mines' positions, two empty spaces were filled with traps that could affect each player separately.

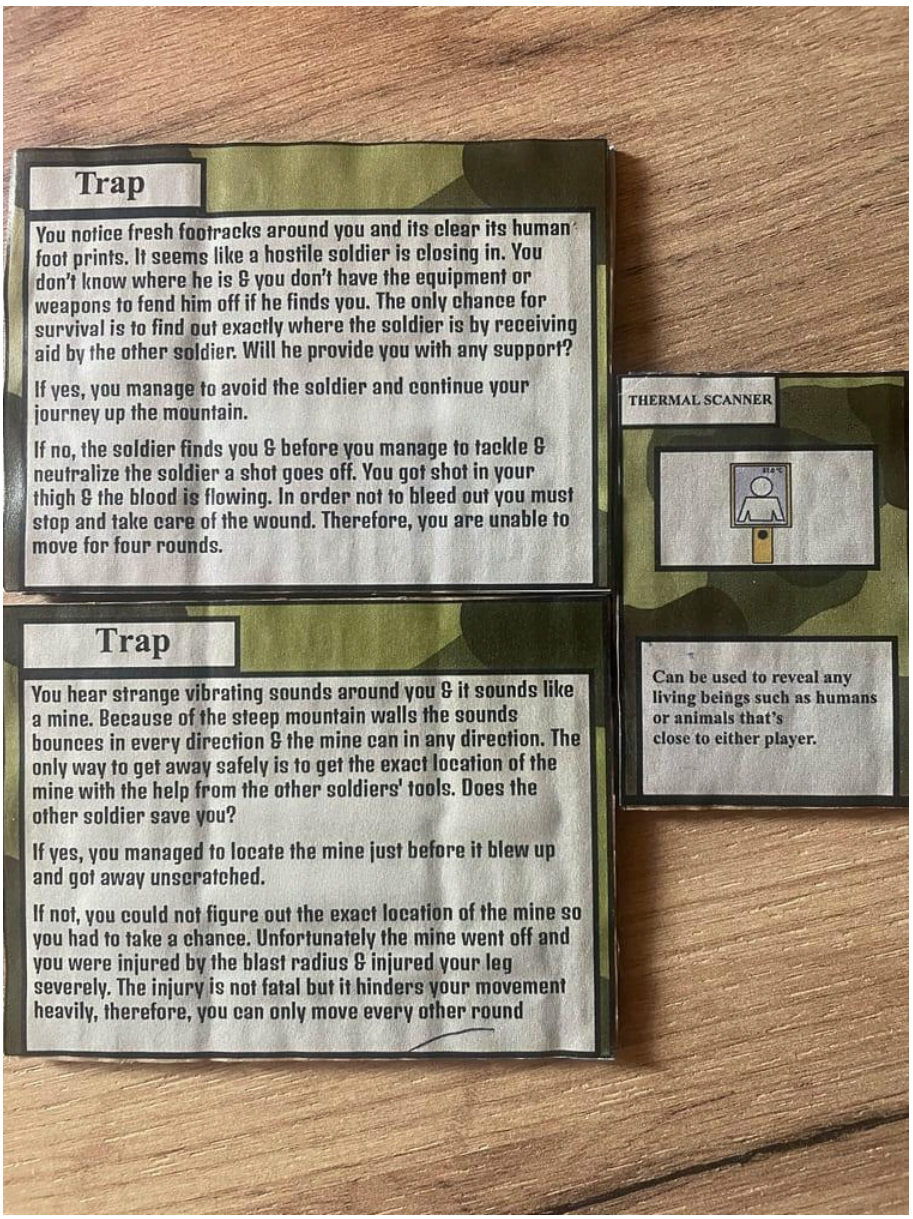


Figure 10: Overview of traps and thermal scanner cards. The image shows two trap cards and a thermal scanner card. Each player has their own trap card, and the other player has to use one of their tools to help. The thermal scanner card was added to the Intel player to serve this purpose.



Figure 11: Trap tokens and backside of trap cards. The image shows the back side of trap tokens and the two trap cards. Trap tokens were added so they could be placed on the game board. The background of the trap cards was numbered to indicate which traps belonged to which player. Number one for the intel player and number two for the tool player.

5.3.2. Design choices prototype 3

The Trap mechanic places a player in a vulnerable position, giving the other player a choice about whether to help. If they choose to help, the player must use one of their resources and free them; if they choose not to help, stepping on a trap would result in a consequence affecting their movement. In the scenario where the intel player steps on a trap, the tool player must use one of their existing tools, ‘Detector’, to help them; if not, the intel player can only move every other round until they reach the end. Whereas when the tool player was stuck on a trap, the intel player was given a new resource called ‘Thermal Scanner’, which can only be used in this scenario; otherwise, the tool player will not be able to move for 4 rounds. Here, the new resource for the intel player and the use of existing tools from the tools

player were implemented to balance the ‘lack of tools’ for intel players, which was mentioned in the previous iteration.

The traps were designed in a way that even though a player steps on both traps, they can be affected only by the first one they step on, giving the other player a chance to experience this mechanic. Moreover, these spaces with traps were represented with question marks, where we intended to create uncertainty about their hostility, allowing players to explore them rather than avoid them, especially since the placements of the traps included the first one being close to one of the players, making them think of stepping on it. The second being in the sixth row, making the second player step on it compulsorily without being affected by the reveal of the first trap. Further, these changes were added to the narrative, trying to align with the story and fiction.

5.3.3. Playtest Session 3

Continuing from playtest session two, the third and final playtest session has one of its main focuses on gathering data about the new trap mechanics and how they affected vulnerability among the players, with a focus on reflective game design. To obtain the data, semi-structured observations and interviews were conducted.

5.3.4. Observations

Main themes that were present in the observation data

1. Dependency between the players

The dependency between the players remains a major presence in this iteration. Players showed greater honesty about their abilities. In the first playtest, the players were more hesitant to reveal their abilities and did not reveal everything early on. However, as the game progressed, they became more open about their abilities.

“I think it’s better if we take the same path, then we can save tools.”

“I have an extra life, so I think you should go where it’s safe.”

The both quotes are from one of the playtests, where the tool player says they were trying to share their resources, depend on the intel player’s tools, and information in order to traverse the mine field and survive. In this iteration, the old detectors and signal jammers were used across both playtests as strategic tools to help both players. Both players were more conscious of sharing the signal jammer and their abilities by carefully strategising and caring for one another, so that each would survive to the end and win together.

2. Vulnerable against the other player

For this iteration, the addition of the traps introduced a degree of vulnerability among the players. The placement of the trap cards on the board proved to be a success since every

player stepped on a trap card. Even though every player helped the other escape the traps, both playtesters felt vulnerable since their fate was in their opponent's hands. If the players decided not to help the other player, they would receive a negative buff, such as limited movement; however, it was expressed that the negative buff is weak and could be stronger, and this could be iterated in future iterations.

3. Risk-taking

During this playtest, the secrecy among the players was less prominent, but the risks they had to take remained high. The players planned where to go and how to use their abilities to benefit both of them. Players discussed how to proceed and who should take which risks. Some players tried to encourage the other to take risks so they did not have to, whereas in another playtest, players risked themselves for others.

“You should be brave and step on it.”

“You want me to take a gamble.”

The above quotes are from the same playtest, at the beginning of the gameplay, the intel player showed slight hesitation in revealing information and told the tool player to take a risk and step on a space. Whereas in the second quote, the tool player is uncertain about the unrevealed space and has expressed that the intel player wants him to take a risk and gamble. After the discussion, both players decided to plan strategically in traversing through mines.

“I can be more risky since I have the signal jammer, I can take more risks. It is a last resort, though.”

In another playtest, the tool player expressed that he could take more risks because he was in possession of the signal jammer so that he could be revived for once. By having the players move simultaneously, the players took risks every round where they decided to move somewhere or do something together since they do not know whether the other player would keep their word or their promises to use the jammer, detectors/detonators, intel, and which steps to take to reach the helicopter at the same time.

4. Uncertainty towards the other player

“You do not give me your full information.”

Even though there was less secrecy during this playtest, the players still showed uncertainty towards the other player's intentions when they shared their abilities. A tool player expressed frustration that the intel player was not giving them full information.

Because they could not directly reveal information about their abilities, the players became uncertain about each other and their true intentions. This mechanic had players taking risks based on the word or promises from the other player, while never truly knowing the other player's intentions.

5. Lack of rewards to betray

The changed win condition for iteration two still brought a constant possibility for betrayal. Even though they did not pursue this path, players saved tools and important information for the very last steps. Even though there was a constant threat of betrayal in the win condition, the game still lacked sufficient rewards to justify betraying the other player, since in the end all players collaborated. If we want to make this aspect stronger, we could introduce more benefits to betray in future iterations.

The players reflected on multiple trust dimensions verbally during the game, including risk, uncertainty, and dependency. The players had the agency to choose how to play, if they wanted to deceive, collaborate and trust each other. The players did not verbally reflect on potential betrayal or vulnerability during the game, but did so in the interview afterwards.

For this iteration, we improved how we modelled vulnerability by adding a trap mechanic, which proved somewhat effective, as the player became directly vulnerable to other players' actions; however, the effect was still limited in creating direct conflict, and vulnerability would need stronger modelling in future iterations, as it also lacked reflection. Dependency, potential betrayal, uncertainty, and risk were modelled successfully, as in the second iteration. For this iteration, all trust dimensions were present in the game, although there is still room for improvement, mainly by making the negative effects of the trap cards stronger and adding more rewards for betraying.

5.3.5. Interview Results

Following a similar approach to previous interviews, after the playtest players were interviewed semi-structurally, and the data from both interviews was analysed using thematic analysis, yielding six major themes that discussed the highlighted dimensions connecting to observations.

Agency to choose your own path (13 instances)

Compared with the previous iteration, the current design resulted in almost twice as many responses from players regarding the freedom to choose their own path. The players' responses explain how flexible the game system is in choosing different ways to traverse the minefield and interact with other players. An intel player said they had the option to share or hide information, and added that it could be natural to hide it at the beginning of gameplay.

“A stranger, I would probably try to collaborate, just because I think it’s a way to get to know someone, then backstab a really good friend.”

One tool player said that, based on the other player in the game, they would play differently: with an unknown player, they would be nice and supportive to get to know each other, whereas with a good friend, they would backstab them and reveal their true play style.

“It was a very clear distinction from the game context and objective context. They can choose whether they want to be more honest or want to be, like, cheating a little bit.”

An intel player said the game varies depending on objectives and game conditions, and they have space to play and choose from. Even though not revealing the information directly falls under the rules, they could still utilise that mechanic and choose to be honest by revealing it or cheat by hiding or manipulating it.

“His or her life is in another one's hands.”

An intel player said the question mark (trap mechanic) means something good that can be used, but they realised it is a trap. So they felt that one player's life was in another player’s hands, which would set the style of the game. Moreover, if they choose to support, it will be cooperative; else, it could be competitive.

“Background information and story makes you adapt to the narrative.”

One of the intel players said that the background of the game characters, the whole introduction, aligns well; all the elements are connected and create a more realistic situation rather than any artificial game content, which further immerses players in the military theme. They said the story has more possibilities, depth, and implications for in-game behaviours.

Dependency and sharing resources (24 instances)

The current iteration yielded a similar number of responses on dependency and resource sharing between players, with comparable levels of exploration.

“I was just like, making sure we both stay alive, and just thinking, how can we both get there at the same time?”

A tool player said they were trying to help themselves and the intel player reach the end and win simultaneously, and said they had no thoughts of making the intel player die. They further added that they tried to understand the intel player’s abilities. When they realised the intel player had more information they could use and rely on, they tried not to use their resources unnecessarily.

“When the exchange of information is not efficient enough, and when I am not familiar with the information, I still need to organise the information and exchange it. So this tension is the main challenge. I would say, during the process”

“So we can figure it out if we exchange information together. And, I think if you offer, another one will offer you too. So it’s just okay”

Both quotes above are from different players and playtests. In the first quote, an intel player stated that the information initially hidden is not received efficiently. However, they still need to organise their own information, but only in a limited way and only for exchange with the

other player. In the other quote, an intel player from another playtest said they believed they could understand the game if they exchanged information, and that if they offered information, the tool player would exchange it with them without issue.

“My experience really felt like collective decision-making thing.”

A tool player said there was effective communication between them and the intel player and liked the intel player for how they came forward with decisions together.

“The jammer is more like a composition to make up for the unclear information.”

An intel player said the signal jammer mechanic can be used to compensate for unclear information. When players do not interact with each other, it can create competition between them, whereas when they interact, it can serve as a resource for one player to share and for another to depend on. Moreover, they added that it could foster collaboration or trust between any two players.

“I think this could be interesting, but it’s a pretty low-stakes collaboration.”

A tool player said the trap mechanic is not very interesting, as it can foster collaboration but feels very low-stakes compared with high-stakes mines. Stepping into a trap space restricts movement for a few turns, not a major effect. It could be interesting if players choose to help each other, and it could affect the game atmosphere, but not the outcome, since one could still win or lose regardless.

Trust reflections (6 instances)

Compared with the previous prototype, players' reflections on trust decreased by two instances but remained noticeable.

“I would say the challenge goes to how you gain the trust of the other player and how you actually build it”

An intel player discussed and reflected on different possibilities of their playstyle. If they were not familiar with the game, they would have eliminated the tool player and gone on to win on their own. However, they were not going to such extremes, as they realised they still needed to talk to the tool player. If they were supposed to go to such extremes, they would advance the game, gain the tool player's trust, and save it to some extent, so they could eliminate the tool player and go to the helicopter (win condition) themselves. They said the biggest challenge in the game could be gaining another player's trust and building it over time.

“The trust game and real life are not so different. The player will keep the impression, the personality and bring them to real life from the game.”

An intel player said the game and real life are connected, with fewer differences in player impressions and personalities, as players will also carry them over into real life. They further said it would be better to play with strangers, as they would not have to worry about making a bad impression.

Deception (22 instances) (Uncertainty and risk taking)

The instances players discussed regarding deception, uncertainty, and risk-taking were similar in number to those in previous prototypes. Here, players discussed feeling uncertain about the other player's possible deception, especially with the use of the can't reveal mechanic.

"I would think maybe he is still trying to hide something from me. So maybe we still have more possibilities, more ability to be used in the game. So I still have some hope."

An intel player noticed that the tool player was hiding information and felt the tool player might have more information and abilities that could be used in the game, so the intel player could still feel a positive side, even though they were not sure about the information and added that the tool player has not specified the number of items they have and has not made it clear in sharing information, which felt very obscure.

"I think it's made this more suspenseful, like, can I trust the other one or not?"

In another playtest, a tool player said that not being able to reveal information made it more suspenseful, as the intel player would be doing the same by hiding it, which created doubt about trusting them. They said this is a nice mechanic, as if a player could read other players' cards, it would simply be a single-player scenario.

Potential betrayal and Testing trust (11 instances)

Even though the current prototype showed a lower response rate than previous ones, their responses are directly linked to the mechanics use case, especially as they discussed the potential of trap mechanics in the possible betrayal.

"It is a collaboration game with some elements where you can also use backstabbing."

While the tool player did not pursue the path of betrayal, this quote reflects on the possibility of doing so.

"Try to make sure the trust from the other players is equivalent to mine, because I need to make sure they are also under the same."

An intel player said they were trying to give information gradually, not everything, and still be as transparent as possible, while keeping something in control, so they can make sure the tool player will not get too angry with them and will not be eliminated accidentally. While doing this, the intel player can test the tool player's trust in them if they were thinking the

same. Moreover, the intel player further added that the tool player said to try to win the game together, but not to get rid of them, which involved testing trust.

“I think the first one is good for people who deceive just to see because neither of them are killed or anything. So try to gain the trust of another person, maybe just to deceive them later”

One of the tool players said the first one, the first trap mechanic that appears in the initial rows of gameplay, gives people who want to use it a good chance to gain another player’s trust and betray them later. Moreover, says that neither of the players would be eliminated by this first trap; rather, a player can be restricted and slowed down for a few moves. Especially if the player has intel cards, they can manipulate by giving false information about the mines, leading the other player into a mine, and win by eliminating the other player.

While an intel player said trap mechanics can be divided into different stages, since they do not share common resources, the first trap could be used to initiate collaboration among players, and this could be an individual decision. However, it is more of a common decision between two players, as there is the possibility of sharing resources. Based on this, the decision towards the second trap is made, and it also creates tensions in the later stage.

5.3.6. Findings of Prototype 3

Main insights and shortcomings

How the design created conditions for players to engage in trust reflection

Based on evaluation, observation, and interview data, it was found that the current prototype created space for players to explore trust on their own and model all trust dimensions. Where dependency through shared resources is most present, the mechanic ‘Can’t Reveal’ created doubts and uncertainty about players' abilities and led players to test each other’s trust in how much they can find and expect in return, making Uncertainty clearly visible between the players. Under the mechanic ‘Can’t Reveal’, players were seen taking risks on each other's verbal information. Lastly, even though betrayal was not seen throughout the playtest, the potential to betray through the win condition, trap mechanic and can’t reveal was discussed by the players, showing that the opportunity to choose betrayal was present, but players tried to avoid it. Player agency has been significantly improved, as reflected in players' reflections on the options available to them, the personalities they can show, and the different play styles they can adopt in the game. Players noted that the new trap mechanics have served as an entry point for collaboration. They said they got the choice to either use it to collaborate or deceive.

What does the current prototype lack to create?

The current prototype was not able to fully model Vulnerability between players, and this lack is a major shortcoming. Vulnerability between the players stayed low throughout all

three iterations. Even though trap mechanics were introduced to create vulnerable situations for players, it became clear that players could not perceive vulnerability in each other. Instead, trap mechanics were considered obstacles that limited player moments rather than being controlled by another player's actions. At the same time, players agreed on the potential of the trap mechanics to utilise it as a base for a deceptive playstyle; they highlighted that if a trap mechanic is used for its purpose, it can place a player's life in the hands of another, introducing vulnerability between the players.

As a result, trap mechanics were not able to generate enough incentives for the players to utilise this mechanic strategically against each other. The lack of meaningful consequences and rewards reduced the motivation to deceive or betray, limiting the emergence of vulnerability. Players have also suggested that the impact of trap mechanics and the incentives associated with them can be improved further.

6. Discussion

In the following section, we will introduce and discuss the main themes gathered and the insights into mechanics from this research through the design project, where we aim to answer our research question: "*How can we create a game system that creates a space for players to explore and reflect on their relationship with trust through the interactions with another player?*"

6.1. Themes

Across all four themes, rather than a separate outcome, reflection was an embedded part of the trust dynamics that the players experienced. This aligns with Khaled's (2018) view, as "Reflection is a process that occurs mentally and appears when individuals encounter situations they cannot address using their previous experiences or solutions". Whereas in reflective game design, reflection is not solely produced by a single mechanic or a moment, but rather by creating spaces for players where they encounter situations that cannot be resolved solely with prior experience but require questioning, evaluation, and reconsideration of choices. Rather than presenting reflection as a separate theme, each of the following themes shows how the game system created conditions for the reflective process to emerge in player interactions.

Theme 1: Potential Betrayal and Testing Trust

"I think the first one is good for people who deceive to just see because neither of them are killed or anything. So try to gain the trust of another person, maybe just to deceive them later"

Here, the tool player is not trying to betray, but rather imagining and considering using the first trap to gain the intel player's trust and later deceive them with the second trap near the end. Even though the tool player pointed out that the traps could not kill a player, they

highlighted possibilities for deception and potential betrayal. According to Khaled (2018), this is exactly what reflective design produces: consideration of different possibilities, not just a fixed way. The fact that none of the players betrayed each other does not undermine the reflection on betrayal; it suggests that players choose to explore the space for reflection mentally rather than act on it.

The findings of this theme indicate that trust is shaped by the possibility of betrayal, which leads players to actively test whether they can trust each other in different situations and to negotiate the exchange of resources between them. Instead of treating trust as an all-or-nothing phenomenon, players evaluated how much trust to give another player as a calculated process. This was made clear by the way players chose to share limited information or resources before deciding to cooperate fully. This kind of behaviour indicates that trust is negotiated and tested before it fully emerges between players. From the first to the third iteration, the potential for betrayal has varied in different ways. Initially, there were no signs of betrayal, whereas testing trust was minimal. In the following iteration two, as per the interview data, both combined were present in 16 instances, and by the third and final iteration, they decreased to 11 instances.

These findings also align with previous research suggesting that, for a player, trust is less dependent on platforms; it emerges through social interactions shaped by resource distribution, competition, deception, negotiation, the possibility of betrayal, and the evaluation of other players' true intentions. Fehlhaber and El-Awad (2024, pp. 29-34) found that, even though elements of distrust were in place, trust emerged between strangers in a competitive multiplayer environment through alliance formation, resource exchange, deception, and dependence among players. This supports the findings of our study, in which trust emerged from interactions among players, shaped by the trust dimensions.

Further, players expressed a need for these exchanges of trust to be reciprocal, to confirm the level of trust they give was balanced. This is supported by Hooman's (1958) work on Social Exchange Theory, in which individuals evaluate costs and rewards in their interactions. The trust between the players became a negotiated, calculated process, in which players sought to ensure their level of trust was matched by the other player. This also aligns with Molm’s (2010) view of reciprocity as a structured process in which players constantly evaluate each other's intentions, and such exchanges influence the emergence of trust.

The win condition, trap mechanic and can’t reveal allowed players to either win on their own, deceive, backstab, or hurt the other player, creating a constant possibility of betrayal. These possibilities created uncertainty, risk and vulnerability for the players, making trust strong because it could be broken.

This kind of reflection, in which players consider various paths but do not act on them, aligns with reflective game design (Khaled, 2018). The game did not need to create betrayal to foster reflection; the potential to betray was sufficient to prompt players to reflect on their decisions and the other player's true intentions.

Theme 2: Sharing Resources and Dependency

"We are soldiers from two different sides, so it feels like you can only benefit yourself by sharing information to survive. Which is a good message all along"

Here, an intel player reflects on the importance of sharing information and relying on each other's resources to survive. The intel player further said the narrative casts the players as enemy soldiers from different sides, but for survival in the game, the benefit comes from sharing information.

"The trust game and real life are not so different. The player will keep the impression, the personality and bring them to real life from the game."

Another intel player is not describing the mechanics above; they are drawing their own conclusions about trust, both in their own lives and in the game. This indicates that the game created a situation where players reflected on trust as a general concept.

The results of this theme suggest that unique and tradable resources create dependencies among players. By having the players depend on each other to succeed, the unique resource mechanics create situations where trust becomes necessary. Sharing resources and dependencies was highly visible from the very beginning. In the first iteration, it occurred in 14 instances; it doubled and remained similar in the next two iterations.

In the game, information sharing was strategic, with players considering it a valuable individual resource, underscoring its significance. Instead of sharing information at once, players have carefully chosen what to reveal and what to hide according to their situation. This suggests that trust is controlled by the sharing of information and resources among players.

Furthermore, players were dependent on each other to achieve their goals, such as surviving, reaching the helicopter together, and reaching the signal jammer, a shared resource. This created a situation in which both players were vulnerable to each other, with each player's success tied to the other's. Therefore, trust became necessary for the players to progress, as going alone was often seen as risky, dangerous, or ineffective.

The Risk observed from choosing an individual path made the players recognise that traversing it could be challenging, with uncertainty about danger and the chances of winning, which, in turn, encouraged them to co-operate and understand the benefits of dependency. This indicates that interdependence can create possibilities for the emergence of trust.

The findings also support Koster's (2018) Trust Spectrum, which states trust as something shaped by design choices such as interdependence, collaboration, information distribution, and the relationship between shared and individual goals. These elements were reflected in our study through mechanisms that created dependence among players, introduced incomplete information, and enabled tension between shared and individual goals. In our

game, players' success relied heavily on dependence and cooperation, indicating that our game lies on the high-trust end of the spectrum.

This perspective is further reinforced by Depping and Mandryk (2017, pp. 451-452), whose research showed that mechanics embedded with cooperation and interdependence can foster social closeness and positive interaction even between strangers. Their findings support the argument that trust can be shaped not only through social behaviour among players but also through mechanics that structure reliance and coordinated action. Together, these studies further reinforce the view that trust can be actively modelled through game design.

Players not only experienced dependency as a necessary aspect of the gameplay, but they also compared their experiences with trust dynamics to real life. This expands the game experience into a broader reflection on trust, allowing players to explore it as a concept in a meaningful way rather than a simple game condition.

Theme 3: Deception, Uncertainty and Risk

"I think it's made this more suspenseful, like, can I trust the other one or not?"

Here, the tool player is not just experiencing suspense or uncertainty; they are asking themselves questions about trust in real time. This aligns with Khaled's (2018) view on the importance of questions over answers, which is foundational to reflective game design.

The findings of this theme indicate that trust emerges through the uncertainty created by the possibility of deception. The trust between the players was shaped by the awareness that the other player could deceive, manipulate, lie, or use them for their own benefit. From the interview data, the potential for deception had consistently increased, starting from 11 to 14 instances in the second iteration. Whereas risk and uncertainty started with seven instances in the first iteration and were minimal in the second iteration of data, but were clearly noted through observations. As for the third and final iteration, all of them combined were 22 instances.

Players have often thought about whether to lie or manipulate the information they gave, yet they have not done so in the end. This highlights that the possibility of deception is enough to influence how players behave and think. In line with Khaled's (2018) work on reflective game design, this suggests that meaning and reflection are created not only by direct actions but also by the possibility of various choices.

The uncertainty between the players often escalated because players had incomplete information to make decisions, or because the other player gave them limited information about their tools or intel. This led the players to make decisions without fully understanding where mines might be or the other player's intentions. This made the players think about how much trust they should give or how much they could trust the other player. Considering the definition of trust by Cho et al. (2015), this reflects the willingness of players to make

risk-based decisions based on subjective beliefs and uncertainty about evidence when receiving rewards.

This uncertainty also led players to be suspicious and doubtful of one another, as they questioned the honesty and reliability of the information they received. Therefore, trust was shaped by doubt and cooperation, which made it appear in different situations. In the context of Social Exchange Theory, this reflects how players weigh potential risks against possible rewards when deciding how to act.

Players have often tried to change how another player understood them by convincing them to trust them. This shows that trust was shaped not only by actions but also by how the players interpreted and communicated them. Therefore, trust became a social process created by the system and its mechanics.

The incomplete information created uncertainty and suspense among the players. It prompted the players to question whether they could believe the other person. The system did not tell players whether to trust or not; it created conditions that forced them to ask themselves whether they should.

Theme 4: Player Agency to make trust decisions

"A stranger, I would probably try to collaborate, just because I think it's a way to get to know someone, then backstab a really good friend"

Here, the tool player discusses a hypothetical situation regarding their own trust behaviour. He reflects that he would behave differently depending on whom he played with. This shows how the game provides players with opportunities to explore it in different ways.

The findings from this theme suggest that player agency is a key factor in enabling the exploration of trust. By allowing players to choose how they want to play and interact with each other, the system and mechanics create a space where players can explore different dimensions of trust. Player agency was present from the first iteration, with seven instances, and gradually increased to 13 in the final iteration.

Players could choose to play honestly or deceptively, giving them control over how trust was developed or undermined. This freedom and agency allowed the players to experiment with different strategies, rather than being constrained by fixed rules. In addition, players could choose to cooperate or compete, thereby shaping the dynamics of trust. Which means that trust was not imposed by the system or its rules, but emerged from the choices made by the players.

Overall, this indicates that agency is essential for reflective game design and reflections around trust. By giving players control over their actions and interactions, the game allows them to explore trust in whatever way they choose. Aligning with Khaled's (2018) work on the importance of allowing players to explore different possibilities.

Across all playtest sessions, vulnerability among the players remained relatively low. Even though mechanics were intended to create it, the final prototype was not able to fully model vulnerability between players. While uncertainty, dependency, risk, and the possibility of betrayal were present, players were rarely placed in situations that could lead to meaningful consequences and rewards, exposing them to the other player's actions. This states that while the game system successfully modelled the dimensions and created conditions for the emergence of reflections regarding trust, it did not fully achieve the level of vulnerability between players.

The results of this theme indicate that players not only experience trust in the game but also reflect on it. Players drew parallels between the game and real-life situations, suggesting that the system and mechanics promoted deeper thinking about trust rather than merely what they had just experienced. From the perspective of principles of reflective game design, these findings align well, encouraging players to reflect on their decisions and their implications. The game system, in this sense, functioned as a space for making various decisions and exploring trust as a concept, rather than simply showing it.

6.2. Mechanics and Design Implications

In this section, we will discuss the mechanics, their modelling as per the trust dimensions, and what went well and what can be improved.

Simultaneous Move

In the first iteration, the simultaneous move modelled uncertainty, since the players had to talk and plan the next moves. Ultimately, they had to trust each other's word about where they decided to go and take a risk, since there was a chance one might go wrong. In combination with the new win condition in the last two iterations, this mechanic modelled vulnerability in the end stages, since the players have to reach the helicopter at the same time. This made the players always vulnerable to one another, potentially leaving one another behind.

This mechanic is modelled towards the theme Deception, Uncertainty and Risk. Players always had to take a risk in trusting one another before making their move, and one could plan to deceive with false information. However, it was observed that players considered deception as minimal to risk and uncertainty.

Unique Resources (Intel and Tools)

The unique resources mechanic, in which players had Intel or Tools, remained successful across all iterations, maintaining a high level of dependency among players, as they were often forced to share resources to succeed. During gameplay, players tried to hide and gradually reveal information, especially at the beginning, to test their trust in one another. This left the players uncertain about the other player's abilities and led them to consider the possibility of being deceived or even betrayed.

These mechanics were most present in the theme Sharing Resources and Dependency. The players had to plan, negotiate, and decide how to use all available resources to benefit both of them in a dangerous situation. This mechanic was successful in modelling dependency between the player throughout all iterations, and it gradually increased with each iteration.

Can’t Reveal

This proved to be one of the successful mechanics, since it modelled four of the five dimensions, with the dependency being the only one not modelled. The can’t reveal mechanic made the players uncertain about the other player’s true intentions and constantly aware of the possibility that the other player could try to deceive/betray them. The mechanic had the players taking risks, as they never had the full information about the other player. The players were constantly vulnerable because they never knew whether the other player was telling the truth. This mechanic is present in all themes except Sharing Resources and Dependency. It made it possible for the players to explore different paths, such as deceiving, testing the other player and betraying.

Signal Jammer

The effect of the jammer mechanic remained the same across all iterations; the main trust dimension the jammer modelled was dependency, as it gave players the option to get an extra life to help and save one another. Therefore, this mechanic was present in the theme Sharing Resources and Dependency. The jammer also enabled the players to take more risks, since they could afford to be eliminated once and continue. Jammer also introduced a level of vulnerability, since the player who owned it controlled its use.

Old Detectors

Throughout gameplay, this mechanic helped establish a dependency between players, since they had to agree to use it together to reveal a space, benefiting both players regardless of their goals. The use of this mechanic varied across iterations. However, it was not central to creating dependency between the players; it was a supporting mechanic to the theme of Sharing Resources and Dependency.

Traps

The traps were the newest iteration of prototype three, and they introduced a level of vulnerability among the players because they could influence each other's outcomes. However, it became clear in the final playtest that the negative effect could be stronger, increasing vulnerability. Since the players could severely restrict the other player's movement and proceed on their own, this mechanic also modelled potential betrayal. Since the players can also help each other, this mechanic models a degree of dependency.

This mechanic is loosely connected to the theme Potential Betrayal and Testing Trust. Potential betrayal was already established in iteration two, and our goal was to implement more vulnerability between the players. However, based on player responses, this mechanic

has flaws and fails to create sufficient vulnerability. The player did talk about the potential this mechanic has if they were to go down the path of betrayal.

Win condition

The change to the win condition in iteration two proved very successful in modelling potential betrayal. In the first iteration, both players could reach the helicopter at different times and still survive, but forcing them to reach it together introduced a constant threat of betrayal, with one potentially left behind at any time. With a high risk of betrayal, it also introduced a level of vulnerability and uncertainty among the players in the later stages of the game. Thus, this mechanic was the main reason the theme Potential Betrayal and Testing Trust was established. Players reflected that they could at any time be left behind by the other player, even if they worked together until the end.

6.3. Answering the Research Question

Regarding our research question "*How can we create a game system that creates a space for players to explore and reflect on their relationship with trust through the interactions with another player?"*, the findings suggest that a game system and its mechanics can support the exploration of trust by modelling all trust dimensions and creating situations in which the level of trust must be negotiated between players rather than assumed. The findings indicate that trust is not an all-or-nothing phenomenon; rather, trust emerges between players through their interactions within systems that introduce dependency, uncertainty, vulnerability, risk and potential betrayal.

This can be achieved by designing mechanics that establish dependencies between players, ensure that cooperation is necessary for progression, allow players to trade unique resources, and meet desired needs by comparing the potential of what they receive with what they give. This closely aligns with social exchange theory (Homans, 1958, p. 606), which posits that rewards outweigh costs, and that the continuity of these exchanges leads to reciprocity (Molm, 2010, para. 1-4) between the players.

Along with dependency, modelling uncertainty from incomplete information about another player's resources requires players to consider each other's true intentions, rather than relying solely on their individual resources and the game system.

Furthermore, introducing the possibility of deception and betrayal is important, as it creates vulnerability and risk between the players, making trust meaningful. Without it, trust becomes pointless and unchallenged. In addition, player agency plays an important role in enabling players to reflect on and explore trust, as the ability to choose among cooperation, deception, and ways of playing allows them to explore different trust dimensions and shape their interactions. This aligns with Khaled's (2018, pp. 20-21) work on reflective game design, in which you need to give the player agency to explore different possibilities on their own, without telling them what is right or wrong.

The system and its mechanics must also place the players in situations where they experience vulnerability and where their outcomes can be affected by another player. Being vulnerable, combined with the risk of being deceived or betrayed, creates situations in which players must make meaningful decisions, which, in turn, shows that trust is experienced through interactions with another player.

Together, these findings indicate that trust can be explored when a game system and its mechanics model the dimensions dependency, uncertainty, risk, vulnerability and betrayal with player agency, allowing trust to emerge as an adaptive and negotiated process, this was further supported by Cho et al., (2015, pp. 7-9) perspective on the importance of modelling the factors affecting the trust within the system, rather than isolating individual aspects.

All these elements connected to the trust dimensions support reflection and exploration by encouraging players to think about their decisions and their consequences, and to meaningfully engage with them, both within the game and in relation to real-world understandings of trust. This highlights the significance of designing the game system so that trust is not assumed but actively explored and constructed through player interaction.

7. Conclusion

This thesis sought to explore how a game system can be designed to allow players to explore trust through its mechanics in an analogue multiplayer game. While trust has been widely studied across multiple disciplines, there is limited research on how specific game mechanics in analogue games can structure and influence players' experiences of trust. To address this gap, this study proposed the research question:

> *How can we create a game system that creates a space for players to explore and reflect on their relationship with trust through the interactions with another player?*

To investigate our research question, we followed a research-through-design approach that involved iterative prototyping and playtesting. Throughout the iterations, we examined how different design aspects shaped player interactions, especially in situations involving uncertainty, risk, vulnerability, betrayal, and dependency. By analysing players' behaviour through observation and their reflections in interviews, we aimed to understand how trust can be modelled and experienced through game mechanics.

Regarding our research question, the findings indicate that trust is not an all-or-nothing phenomenon but emerges through mechanisms and player interactions in situations characterised by uncertainty, vulnerability, potential betrayal, dependency, and risk. The level of trust between the players had to be negotiated rather than assumed, especially in situations where information was incomplete and the possibility of deception was present. These elements, combined with player agency, enable players to meaningfully engage and reflect on trust as part of the gameplay.

This study contributes to understanding how trust can be modelled in a game system, with a focus on exploring and creating a reflective experience. By showing how specific mechanics can model different dimensions of trust, our research could help and inspire other game designers seeking to create games about complex concepts.

Takeaway 1 - Trust requires all five dimensions together

Trust did not emerge from a single mechanic or dimension, but from the interaction between dependency, uncertainty, risk, vulnerability, and potential betrayal. Together, these dimensions created situations where players had to actively negotiate, question, and reflect on trust rather than assume it.

Takeaway 2 - Restricting information is more powerful than restricting action

The “Can’t Reveal” mechanic modelled multiple trust dimensions simultaneously, making it the most efficient mechanic in this study. By limiting what players could communicate rather than what they could physically share, the mechanic encouraged players to interpret each other's intentions, question honesty, and reflect upon how trust was negotiated under uncertainty. Designers seeking to create trust dynamics may therefore benefit more from restricting information between players than from restricting actions.

Takeaway 3 - The possibility of betrayal matters more than betrayal itself.

None of the players chose to betray, but potential betrayal was one of the most discussed and reflected-upon aspects across the last two iterations. The possibility of betrayal created tension, uncertainty, and continuous evaluation of the other player’s true intentions. This suggests that creating situations where betrayal could occur may promote more reflection than mechanics that directly force betrayal.

Takeaway 4 - Win conditions shape trust more than most mechanics.

Changing the win condition to require players to reach the helicopter together had the greatest impact on player behaviour, cooperation, and reflections. This changed how players approached information sharing and negotiation, demonstrating that win conditions can serve not only as end goals but also as mechanics that shape how players experience and reflect on trust.

Takeaway 5 - Vulnerability is the hardest dimension to model.

Across all three iterations, vulnerability among players was the weakest dimension in the game system. While the system made players experience uncertainty and dependency, it could not place them in situations with meaningful personal consequences. This suggests that designing for vulnerability requires carefully balanced stakes that are significant enough to model vulnerability between the players, thereby encouraging emotional investment and reflection while still allowing players to feel willing to engage with the system.

We believe our research could be useful for game designers interested in creating games that promote reflection on complex social concepts and relationships by showing how a social concept like trust can be explored through a game system and its mechanics, rather than solely through narrative. We also believe our game can be used in educational environments, workplaces, or team-building contexts where the goal is to promote conversations and reflections on cooperation, communication, relationships, and trust as a general concept.

As designers, this project helped us recognise the impact of intentional design choices and iterative playtesting while working closely with complex social concepts. It also deepened our understanding of how mechanics not only impact player behaviour but also shape player reflection. Through continuous playtests, iterations, and our dedication to the game, we developed a better understanding of trust, communication, and collaboration within the design process itself, which further reinforced our view of game design as both a creative and reflective process.

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10. Appendices

10.1. Consent form

CONSENT TO PARTICIPATE IN RESEARCH

Researching how game systems and mechanics impacts social interaction

Principal Investigator: Marcus Hägg and Venkata Vamsi Krishna Kola

Institution: Uppsala University, Campus Gotland. Department of Game Design

What is the purpose of this research? We are asking you to be in a research study because we are trying to learn more about how mechanics can potentially influence the social interaction between two individuals. This research is being conducted by Marcus Hägg and Venkata Vamsi Krishna Kola at Uppsala University. Our goal is to include about 11 participants in the research.

What is involved in being in the research study? If you agree to be in this study, you will be asked to play our game while being observed by the researchers and you will be part of an interview after you have finished the game which all together will take approximately 90 minutes. You will be asked basic questions such as your age, who you are, your previous experiences related to games and your experience with playing our game. The interview will be recorded in order for the researchers to analyse and transcribe into written notes later on. The researchers will observe you and write down notes when you play. The researcher will let you know when the recording starts.

How much time will this take? This study will take about 90 minutes of your time in total. The gameplay will take approximately 45 minutes and the interview will take approximately 30 minutes.

Are there any risks involved in participating in this study? There are not any risks involved by being a part of this study. You may feel uncomfortable about answering certain questions. You do not have to answer any questions you do not want to. You have the option to stop playing the game or interview at any time. Our research will be published in Diva so there is a possibility that others may find out what you have said in the interviews, however, security measures have been put in place to protect your integrity and prevent that from happening.

Are there any benefits to participating in this study? You will not receive any personal benefits from being a part of this study. The goal with this research is to get a better understanding of how mechanics potentially can influence the social interaction between the players.

Are there any costs to me for being in the research? No.

Can you decide not to participate? You can choose not to participate at any time and you can choose to withdraw from the research if you change your mind. There will not be any negative consequences or penalties if you choose not to participate.

Who will see my study information and how will the confidentiality of the information collected for the research be protected? The research recordings and notes will be stored securely on an external hard drive. Your information will be combined with other information from other participants taking part in the study. We will not include your name or any other information that could directly identify you. Instead of using your name, you will be assigned a code name to de-identify you, for example, participant three. All data will be destroyed after this study is complete.

Who should be contacted for more information about the research? Before you accept to be a part of this study, please read through this document thoroughly and ask any questions that come to mind. If you have any questions, concerns, complaints or want more

information about the study, you can contact the researcher Marcus Hägg directly at xxx during business hours on week days, 08-16. You will receive a copy of this consent form.

Statement of Consent from the Subject:

□ I authorize the researchers Marcus Hägg, Venkata Vamsi Krishna Kola and Uppsala University to take notes and audio recordings of me during interviews in connection with this research study. The data will be destroyed after the study is complete.

□ I authorize the researchers Marcus Hägg, Venkata Vamsi Krishna Kola and Uppsala University to use the collected data for our paper and to publish it.

Signature: _____ Provider Version

Printed Name: _____

I have read the above information. I have had all my questions and concerns answered. By signing below, I indicate my consent to be in the research.

Signature:_____

Printed name: _____

Date: _____

10.2. Observations

Goal - What should be observed

The observation will focus on behaviors related to trust in regards to the game mechanics and system. We will observe behaviours connected to:

Risk taking - Willingly take risks based on another player's word.

Vulnerability - Resource/information sharing

Uncertainty - How players navigate not being able to show their information directly

Betrayal - Deception, lies or broken agreements.

Reflection - Their reflection based on their decision related to trust.

Observation Protocol

1. Playtest information - Date, playtest number, observer, game version (which iteration). Does the player know each other? (Try to avoid)
2. Observation structure - Mechanics involved (which mechanic enabled that situation), player action, trust dimension, relevant dialogue, outcome.
3. Reflection Indicators - During the playtest look for if players verbally reflect on trust, do they comment on regret/betrayal, do they refer to the mechanics when explaining

How the observation will be recorded

The observation will be recorded by taking digital notes during the playtest and we will use a semi structured observation approach, where we will have a set of behaviours we will look for but still leave room for anything unexpected.

For each trust related event, we will follow the observation protocol which includes.

1. Mechanic involved
2. Player action
3. Trust dimension
4. Relevant dialogue
5. Outcome

How to ensure accuracy of observation

Using the semi structured approach while having a pre defined set of categories to observe for (Trust dimensions). We will only focus on behaviours we directly observe in order to avoid any assumptions. We will conduct a pilot test to find flaws in the structure of our observations and adjust them before the data collection starts. We will remain neutral, let the player play rather than giving instructions.

Relationship between the observers and participants

During the playtest the observation will be:

Non-participatory, which means that we as the observers will not interfere with the gameplay and observe from a distance.

Overt, since we will make the participants aware that they're being observed.

10.3. Semi structured interviews

1. How would they explain the game and what it's about to someone who has not played it?
2. How would you describe in broad strokes your experience of playing the game?
3. Are there any moments that created doubt? If so, can you explain those moments?
4. When you traversed the mine field, how did that feel to you without knowing the exact location of the mines and other player's resources?
5. What challenges did you face during the game and how did you try to overcome them?
6. Can you speak to how you related to the other player throughout the game? (how did you try to interact with the other player?)
7. How did you feel about not being able to reveal the information directly?
8. What did you think about the extra life?
9. What stood out the most to you when playing this game?
10. Did the experience remind you of any situations outside the game?
11. What do you think the game was about?
12. If you had to tell a friend about your experience with the other player during the game, how would you describe that person?

10.4. Rules and Narrative introduction

Rules

Your objective is to make it to the top of the mountain where the rescue helicopter is waiting without getting eliminated by a mine. If you step on a mine, you will be killed and eliminated from the game; however, the other soldiers may still continue. Because of the time constraints and the dangers of staying on land for too long, the helicopter will take off as soon as one soldier reaches it. Therefore, in order for both soldiers to survive, they need to reach the helicopter simultaneously. Both of you move one step simultaneously each round, and you

may not return to your starting position after making your first move; you can move however you want.

1. When you step on a space, flip over the token to check if it contains a mine and remove the token from the board.
2. If you step on a mine, you die and are eliminated from the game. If there is no mine, you're safe.
3. If both of you step on the same mine simultaneously, you are both eliminated.
4. Each player has a unique set of abilities that can be used to their advantage and is given a set of cards and a rule sheet describing their abilities. At the start of the game, each player selects their abilities from either set A or B. Players may not switch abilities after choosing one.
5. There is a signal jammer placed near the centre of the board, and players collect it by landing on that space. The signal jammer can only be collected by one player, and if both land on that space simultaneously, the players have to decide which receives it. If the players can't decide who gets the jammer, it stays where it is. The signal jammer provides the ability to block a mine from going off if a player steps on a mine, thereby surviving stepping on a mine. The signal jammer can only be used once.
6. Both players have an old and beat-up mine detector, which can be used by both players together to safely reveal a location on the map by flipping over a token that they both agree on. The detectors can only be used once, and players must agree to use them together. The old detectors cannot be used individually.
7. If a player steps on a token with a question mark, said player picks up their face-down question mark card and reads it out loud.
8. If a player has already stepped on a question mark token and read their card, they will not be affected if they step on the other question mark token, and the player should place their player token on top of it.
9. Since the soldiers are from enemy nations and communicating through their radios, the players cannot show any of their personal ability cards to each other; this does not include the beat-up detectors and signal jammer card. The players can, however, communicate freely about their abilities.

Narrative introduction

Act of Survival is a two-player synchronous strategy game.

Two enemy soldiers from different nations have found themselves as the only ones who have managed to survive the war from their respective units and are now stranded in this treacherous warzone.

The soldiers have managed to find the same frequency on their radios, and therefore they can communicate with each other and transmit their locations. Both soldiers heard a vague distress call from a search and rescue helicopter from an unknown nation that was scanning the area for survivors. They manage to transmit their location and the rescue party communicates to them that time is scarce and hostile soldiers are still roaming the area

looking to finish off any survivors, therefore the helicopter will not be able land for long, the extraction must happen quickly and the only area to safely land is on top of the mountain which is covered with dense forests and sensor mines are scattered all over the mountain.

The mountain is full of dense vegetation; therefore, spotting the mines with only eyesight is impossible. One of the soldiers has managed to contact his headquarters through his radio, and they have satellites that can search the area for any signs of where the mines are located. However, because of the density and environment of the mountain, the intel he receives about the location of the mines is constantly interfered with, and only small parts of the intel make their way through. The other soldier went back to his unit's camp and managed to salvage whatever equipment that was not destroyed in order to help him deal with the dangers of the mountain, but he did not manage to salvage nearly enough to clear a safe path over the whole mountain.

On the mountain, there is some equipment that can be salvaged from the war that the soldiers can find and utilise. A signal jammer that can block the sensor of the mine in case a soldier comes too close to a mine. There is also a pair of beat-up old mine detectors for them to use; however, because of the condition, the soldiers know it is not wise to trust the information of one detector, instead using them both to confirm a location is safe.

Will they manage to survive the mountain and reach the rescue helicopter in time?

10.5. Observation and Interview Data

Act of Survival - Observation Data

Data Coding_Playtest session 1

Data Coding_Playtest session 2

Data Coding_Playtest session 3

10.6. Gameplay Video

Gameplay video.mov